

Senior High School



Lesson Exemplar in General Mathematics

Quarter 1

Unit

2

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Lesson Exemplar for General Mathematics
Quarter 1: Unit 1

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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR					
Learning Area:	GENERAL MATHEMATICS				Grade Level: 11
Semester:	FIRST	Quarter:	FIRST	Unit:	2

I. OBJECTIVES	
Content Standard	The learners demonstrate knowledge and understanding of patterns, arithmetic and geometric sequences.
Performance Standard	The learners are able to describe and present how they use sequences and series to make informed decisions on real-life situations involving earning money, buying and selling, acquiring loans, making investments, and planning purchases.
Learning Competencies	The learners... 1. describe patterns by inspection including patterns in art and nature. 2. determine the next term of a given pattern, including Fibonacci sequence. 3. identify the rule of a given pattern 4. illustrate the attributes of arithmetic and geometric sequences. 5. solve problems involving arithmetic and geometric sequences. 6. differentiate sequences from series. 7. illustrate arithmetic and geometric series. 8. represent series using sigma notation and vice versa 9. find the sum of arithmetic and geometric series. 10. solve problems involving arithmetic and geometric series.

II. REFERENCES and MATERIALS	Online Sources Cuemath. (n.d.). <i>Sequence and series</i>. https://www.cuemath.com/numbers/sequence-and-series/ Mathcentre. (2009). <i>Arithmetic and geometric progressions</i>. https://www.mathcentre.ac.uk/resources/uploaded/mc-ty-apgp-2009-1.pdf Mirzaagha, M. (n.d.). <i>Sequences and series</i>. California State University, Sacramento. https://www.csus.edu/indiv/m/mirzaagham/math1/SQ6.pdf Phillips, C. (n.d.). <i>Unit 8 day 1 notes – Geometric sequences</i>. https://coachchadphillips.weebly.com/uploads/1/2/0/8/120840490/unit_8_day_1_notes_-_geometric_sequences.pdf Roberts, C. (n.d.). <i>Cambridge math studies book</i>. https://robertsmathclass.weebly.com/uploads/6/4/0/3/64033291/cambridge-math-studies-book-_1_-1-346-1-153-79-153.pdf Cooper, J. (n.d.). <i>Arithmetic and geometric series</i> (Chapter 4). https://coopersnotes.net/docs/elementary%20algebra/CHAP04%20Arithmetic%20and%20Geometric%20Series.pdf Macquarie University. (n.d.). <i>Module 3: Worksheet 36</i> [Numeracy Project]. http://maths.mq.edu.au/numeracy/web_mums/module3/Worksheet36/module3.pdf Video lessons Sequences and Series https://www.youtube.com/watch?v=REDVjUTvXvk&t=368s Types of Sequences https://www.youtube.com/shorts/qW0yF2_-LEc Arithmetic Sequence Link: https://www.youtube.com/watch?v=jExpsJTU9o8 Arithmetic Series Link: https://www.youtube.com/watch?v=GZH68SubgRE Geometric Sequence
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	Link: https://www.youtube.com/watch?v=DVpfJC-waw&t=76s Geometric Series Link: https://www.youtube.com/watch?v=kygPMFN4baM Materials • Calculators • Visual aids/PowerPoint presentation • Worksheets • Task cards • Budget Template	
III. CONTENT	Lesson 2.1: Sequences <i>Recommended Time: 2 hours</i>	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. identify patterns found in art and nature. 2. describe a sequence. 3. classify sequences as finite or infinite based on their structure and number of terms. 4. determine the next term of a given sequence, including a Fibonacci sequence. 5. describe the rule for finding the next term in a sequence	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATIONS (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1.1: And so... Exploring Animal Life Cycles Instructions 1. The class will be divided into small groups. 2. Each group will watch a documentary video showcasing an animal's life cycle (for example: the transformation of caterpillar into a butterfly) 3. While watching, take note and/or sketch each stage of the animal life's cycle. Pay attention to the sequence, pattern and characteristics of each stage.	Activity A.1.1: And so... Exploring Animal Life Cycles This activity refreshes basic math operations and pattern recognition. To facilitate it, the teacher may apply one of the following strategies: Strategy 1: Collaborative Work 1. Begin by explaining the objective: to understand sequencing by observing the stages of an animal's life cycle.

4. After the video, collaborate with your group to create a visual presentation—such as a poster, flipbook, or digital slideshow—illustrating the sequential stages of the observed life cycle.
5. Present your group’s output to the class.

Activity A.1.2:
What’s Next? Replace with:

Instructions

1. The class will be divided into three groups.
2. Each group will be given a sequence and asked to extend it by three terms.
3. A representative from each group will write the next three terms on the board.
4. The group will explain to the class how they determined the next three terms.
5. The class will decide whether the group’s answer is correct.

Sequences:

Group 1: 3, 6, 12, 24, _____, _____, _____

Group 2: 0, 1, 1, 2, 3, 5, _____, _____, _____

Group 3: 1, 3, 6, 10, _____, _____, _____

Answers:

Group 1: 48, 96, 192

Group 2: 8, 13, 21

Group 3: 15, 21, 28

2. Show a documentary video depicting the life cycle of a specific animal. Encourage students to watch attentively, noting the sequence of stages.
3. Introduce sequencing words such as "first," "next," "then," and "finally." Discuss how these words help describe the progression of events or stages.
4. Divide students into small groups. Instruct each group to create a visual representation (e.g., poster or storyboard) of the animal’s life cycle, using sequencing words to label each stage. Encourage creativity.
5. Have each group present their visual representation to the class. Facilitate discussion by asking questions such as:

What did you learn about the animal’s life cycle?
How did using sequencing words help you understand the progression?

You may visit the following links for videos:

<https://www.youtube.com/watch?v=G3VMl-EY2DM>

https://www.youtube.com/watch?v=ws_D5nXOAJg

Strategy 2: Group Work

1. Provide necessary materials: printed sequences with missing terms and a pen.
2. Explain the objective: “Before describing what a sequence is, analyze the pattern of the given numbers and then extend the sequence by continuing the pattern.”
3. Present instructions:
 - a. Learners analyze the pattern of the given numbers.
 - b. They discuss among themselves the next three numbers.
 - c. The group shares answers and reasoning with the class through a representative.
4. Facilitate discussion by briefly reviewing why each skill is important for the lesson.

Strategy 3: Brainstorming

		1. Provide the class with the three sets of numbers. 2. Have the whole class analyze each set to determine the pattern and find the next terms. 3. Poses guiding questions to help students reason through the sequences: • “Is there any pattern that you can see?” • “What operation is being applied?” • “How do you get the next number?” 4. Let the class decide if the next numbers are correct to encourage discussion and foster critical thinking. Strategy 4: Race for the Will Begin the activity by telling a story: “A rich old man has left a number code in his will. Whoever among his three children can first crack the pattern and figure out the seven numbers to complete the code will inherit his wealth. The codes follow a special order called a sequence. Let us see who among his children can solve the code first.” The numbers are: 1, 1, 2, 3, 5, ____, ____ 1. The teacher may group students and turn this into a game. 2. Let the class decide if the next numbers are correct to encourage discussion and foster critical thinking.
	<i>A.2. Establishing the Purpose of the Lesson</i> Activity A.2: Pattern Hunt Instructions: 1. Pictures from nature and artwork featuring striking images that highlights patterns will be shown through a slide show. 2. Observe the pictures and look for repetition, symmetry, shapes, or structure. 3. Discuss your observations with a partner, then share your insights with the class.	Activity A.2: Pattern Hunt In this activity, students will observe a slideshow of images from nature and art that display clear patterns such as repetition, symmetry, shapes, and structure. They will identify and discuss these patterns with a partner and the class. This activity introduces the concept of patterns through real-life examples and serves as a foundation for lessons on recognizing, extending, and representing patterns in mathematics. To facilitate this activity, the teacher may apply one of the following strategies:

Nature:



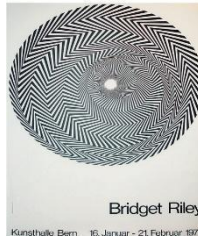
Art:



Wassily Kandinsky, Circles in a circle, 1923



Frank Stella's 'Haran II', 1967



Bridget Riley
Kunsthaus, Bern 10. Januar - 21. Februar 1971

Strategy 1: Hear Me

Instructions:

1. Prepare a slideshow featuring various pictures captured from nature and art.
2. Let students observe the pictures carefully, noting details such as symmetry, repetition, shapes, and structure.
3. Have them discuss their observations with a partner, then share their insights with the class.
4. Connect to the lesson by asking students to give more examples of patterns they see in nature.
5. After sharing, say:

"Just like we see patterns in nature and art, we can find patterns in numbers too. These number patterns are called sequences. Today, we will describe sequences of numbers through observation, inspecting what changes and what repeats—just like what we did with the images."

Strategy 2: Pattern Gallery Walk

Instructions:

1. Post printed images from nature and art around the room.
2. Divide students into small groups and have them visit the posted images.
3. Ask them to describe the patterns they see using terms like repeating, increasing, decreasing, and symmetry.
4. Connect to the lesson by asking students to give more examples of patterns they observe in nature.
5. After sharing, say:

"Just like we see patterns in nature and art, we can find patterns in numbers too. These number patterns are called sequences. Today, we will describe sequences of numbers through observation, inspecting what changes and what repeats—just like what we did with the images."

		Strategy 3. Pattern Hunt Instructions: 1. Divide students into small groups and have them search for real-world patterns (shapes, colors, tiles, etc.) around the campus. 2. Let them capture photos of the patterns they find using their phones. 3. Have them present and describe their findings to the class. (Optional: Students may compile their photos into a digital “Pattern Museum” or a class poster.) 3. Connect to the lesson by asking students to give more examples of patterns they see in nature. 4. After sharing, say: <i>“Just like we see patterns in nature and art, we can find patterns in numbers too. These number patterns are called sequences. Today, we will describe sequences of numbers through observation, inspecting what changes and what repeats—just like what we did with the images.”</i>
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Pattern Please Materials: Printed strips with different types of sequences. Examples of sequences on strips: 4, 8, 12, __, __ (add 4) 1, 2, 4, 8, __, __ (multiply by 2) 1, 1, 2, 3, 5, __, __ (each term is the sum of the two preceding terms) 1, 4, 9, 16, 25, . . . (squares) 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, . . . (reciprocals) 1, $\sqrt{2}$, $\sqrt{3}$, 2, $\sqrt{5}$, ... (square roots)	To facilitate this activity the teacher may apply one of the following strategies. Strategy 1: Guided Group Discussion Instructions: 1. Introduce the activity by saying: <i>“If nature has patterns, so do numbers. Let us now explore some examples of number sequences.”</i> 2. Hand out the strips to pairs of students. 3. Ask students to identify the pattern, continue the sequence, and explain their thinking. 4. Have them identify the rule they discovered and share it with the class. 5. Emphasize how to determine the next terms in the given sequences. Discussion Prompts: • What is changing between the terms? • How do you know the rule works? • Can you predict the 6th term?

B.2. Discussing the Concept

Activity B.2: Lesson Discussion of Sequences

Instructions:

Study the information and examples and prepare for class discussion.

Key Terms and Definitions

Sequence - a list of numbers arranged in a specific order, following a rule.

Example: 3, 6, 12, 24, . . .

Term - each number in a sequence.

Example: In the sequence 3, 6, 12, 24, each of the numbers 3, 6, 12, and 24 is a term.

Position (or Index) - the place of a term in the sequence, usually counted as 1st term, 2nd term, and so on

Notation: a_1 is the first term, a_2 is the second term, etc. In general, a_n denotes the n th term

Rule (or Pattern Rule) - a description of how the terms is formed or how they change from one to the next.

Example: "Add 2" in the sequence 2, 4, 6, 8, ...

Fibonacci Sequence - a sequence where each term after the first two is the sum of the two preceding terms.

Example: 1, 1, 2, 3, 5, 8, 13, ...

Classification of Sequences

(based on the number of terms)

Finite Sequence

A finite sequence has a specific number of terms – it has a beginning and an end.

Activity B.2: Lesson Discussion

In this part, students will study key terms and examples related to sequences in preparation for a guided class discussion. They will learn essential vocabulary such as term, position, rule, and notation, as well as recognize different types of sequences including finite, infinite, and the Fibonacci sequence. This activity builds the foundational understanding needed to analyze, extend, and represent sequences mathematically, forming a crucial step toward deeper pattern recognition and generalization in succeeding lessons.

To facilitate the discussion, the teacher may select any of the given strategies or any strategy that suits the needs of the learners. Additional examples may be provided if necessary

Strategy 1: Interactive Formula Mapping

Instructions:

1. Prepare sets of cards:

Set A: Key terms

Set B: Definitions

Set C: Examples

2. Distribute shuffled cards to small groups.

3. Groups must correctly match all three: term, definition, and example.

4. After the given time, review answers as a class.

Strategy 2: Word Hunt

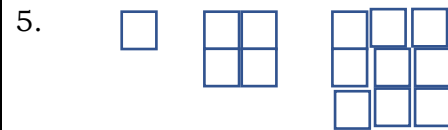
Instructions:

1. Prepare cards with terms and definitions written on them.

	You can count all the terms. It stops after a certain point. <i>Example:</i> 2, 4, 6, 8, 10 (This sequence has 5 terms – it ends at 10) Infinite Sequence An infinite sequence has no end- it goes on forever. It keeps following the same rule endlessly. You can describe the rule, but you can't list all the terms. <i>Example:</i> 1, 2, 4, 8, 16, 32, ... (This keeps doubling and never stops.) In mathematical notation, an infinite sequence often includes ellipsis “...” to show it continues.	2. Place the cards around the room or distribute them randomly. 3. Divide students into small teams. 4. Each team searches for cards that match terms with their corresponding definitions. When they find a correct pair, they return to the teacher for confirmation. 5. After all cards are matched, discuss each pair with the class, ensuring everyone understands how the definitions and terms connect. 6. Practice with examples using the definitions. Strategy 3: Peer Teaching Instructions: 1. Create teaching stations for each term with its definition. 2. At each station, prepare: The definition for that term. An example related to the term. 3. Split the class into small groups and assign each group to a station. 4. Each group explains the term, definition, and example at their station. 5. After the given time, rotate the groups to the next station. 6. When all groups have visited each station, bring the class together to discuss any challenges or findings. Strategy 4: Video Lesson To reinforce student understanding of sequences and series, you may integrate the following video material into your lesson: Title: Sequences and Series Link: https://www.youtube.com/shorts/qW0yF2_-LEc How to Use:
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		1. Download or stream the video ahead of time to avoid connectivity issues in class. 2. The video provides a step-by-step explanation of the concepts with examples and visuals, ideal for learners needing reinforcement. 3. Play the video during the session, pausing at key points to explain or clarify. 4. After the video, engage students in practice problems to apply what they learned in a hands-on, guided way.
	B.3. Developing Mastery Activity B.3: Practice Problems Instructions: Do the following: a. Give the next term of the given sequences. b. Identify the rule. 1. 81, 27, 9, 3, ____ Answer: a. 1 ii. divide the previous term by 3 to get the next term 2.  Answer: a. 10 b. each term is the sum of the two previous term 3. 7, 14, 28, 56, ____ Answer: a. 112 b. multiply by 2 4. 10, 21, 31, ____ Answer:	Activity B.3: Practice Problems In this activity learners will firm-up their learnings by answering another set of problems. Teacher may apply any of the following strategies. Additional items may be provided. Think-Pair-Solve-Share Instructions: 1. Give each student the set of problems. 2. Ask them to solve one problem alone (Think). 3. Then pair them with a seatmate to discuss and solve the next problem together (Pair + Solve). 4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). 5. Use visual aids like posters or digital slides to display each problem. Color-Coded Stations Instructions: 1. Prepare 5 index cards or printed task cards, each with one problem (use different colors for each type of problem). 2. Divide the class into small groups of 4–5 students. 3. Give each group one card to solve together. 4. After the allotted time, say “Pass!” and have groups pass their card clockwise to the next group.

- a. 52
b. the next term is the sum of the two term before it



Answer:

- a. 16
b. Squares of whole numbers

6. $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, ____

Answer:

- a. $\frac{1}{16}$
b. getting half

7. 8, 27, 64, ____

Answer:

- a. 125
b. cubes of whole numbers

8. 3, 6, 10, 15, ____

Answer:

- a. 21
b. triangular numbers

9. 1×2 , 2×3 , 3×4 , ____

Answer:

- a. 4×5
b. each term multiplies a number n by the next consecutive number, $n+1$

10. 5, 10, 15, 20, ____

Answer:

- a. 25
b. multiples of 5, or add 5

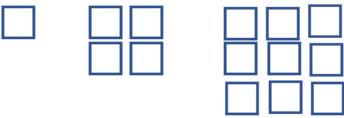
5. Continue until all groups have solved all cards.
6. Conduct a quick review afterward to clarify steps and encourage reflection.

Interactive Whiteboard Relay

Instructions:

1. Divide the class into two teams.
2. Display one problem at a time on the board or projector.
3. A student from each team races to solve and explain their answer on the board.
4. Award points for correct answers and clear explanations.
5. Encourage class discussion after each round—ask if anyone solved it differently or spotted errors.

C. Demonstrating Knowledge and Skills	<i>C.1. Finding Practical Application</i> Activity C.1: Saving Up Smartly – A Real-Life Application of Sequences Scenario: A student wants to build a personal savings fund. They commit to saving ₱50.00 in the first week and increase the amount they save by ₱15.00 each week. Week 1: ₱50.00 Week 2: ₱65.00 Week 3: ₱80.00 ... (This continues as an arithmetic sequence.) <i>Task:</i> Use your knowledge of sequences and series to analyze the student's savings plan. Then answer the following questions: 1. How much will the student save in Week 10? 2. How much will the student have saved in total after 10 weeks? 3. If the student's goal is to save ₱1,500.00, in how many weeks will the goal be met or exceeded? 4. How can this type of savings plan help in achieving financial goals such as buying a gadget, funding a project, or starting a small business? (Short reflective question) <i>Expected Answers:</i> 1. ₱185.00 2. ₱1,175.00 3. 12 weeks 4. Student-generated responses; should reflect the importance of planning and consistent saving.	Activity C.1: Saving Up Smartly – A Real-Life Application of Sequences This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, preparing them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. In this part you will apply Experiential-Collaborative learning.
	C.2. Making Generalization	Activity C.2: Patterns Speak

	Activity C.2: Patterns Speak Instructions: Students will identify and express general truths about sequences. 3, 6, 9, 12, 15  100, 50, 25, 12.5 Questions: 1. What is changing in each sequence? 2. How is each term related to the previous term? 3. What rule can you generalize from each sequence?	To help students make meaningful generalizations, the teacher may use the Think-Pair-Share approach combined with sentence frames to promote reflection, discussion, and synthesis for deeper understanding. Steps: 1. Think (Individual Reflection) Let the student write his/her reflection on the question: “What patterns or rules did you notice in the sequences we studied today?” 2. Pair (Collaboration) Sharing thoughts with a partner. Prompt: “Compare what you noticed in the lesson. Did you find the same rule? What general statement can you form?” 3. Share (Class Synthesis) Invite at least three student volunteers to share their general statements with the class. Write the shared generalizations on the board. Sequence Storytelling 1. Assign each group a sequence. 2. Groups create a short story or scenario where their assigned sequence is evident. 3. Classmates identify the sequence and discuss its characteristics.
	C.3. Evaluating Learning Activity C.3: Assessing Level of Understanding Instructions: Read each item carefully and write the correct answer. 1. What is true about the sequence represented by the pattern below? 	Activity C.3: Assessing Level of Understanding This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation. Additional items may be provided.

- a. Each term in the sequence is the sum of all counting numbers from 1 up to that term's position in the sequence
- b. Each term in the sequence is the square of all counting numbers from 1 up to that term's position in the sequence
- c. Each term in the sequence is the product of all numbers from 1 up to that term's position in the sequence
- d. Each term in the sequence is the thrice the sum of all counting numbers from 1 up to that term's position in the sequence

Answer: a

2. What is the next term in the sequence: 1, 1, 2, 3, 5?

- a. 6
- b. 7
- c. 8
- d. 10

Answer: c

3. What is the rule in the sequence: 3, 6, 12, 24, ___?

- a. Add 3
- b. Add 6
- c. Multiply by 2
- d. Multiply by 3

Answer: c

4. A sequence has a pattern: add 4 each time. The first term is 2, what is the 5th term?

- a. 10
- b. 18
- c. 20
- d. 22

Answer: b

	5. A group of students is arranging chairs in a theater where each row has 3 more chairs than the previous one. The first row has 10 chairs. How many chairs are there in the fifth row? a. 13 b. 16 c. 19 d. 22 Answer: d	
	C.4. Additional Activities Activity C.4: More Exercises on Sequence Remediation Instruction Determine the next term of the sequence and give its rule. Identify the type of sequence. 10, 7, 4, 1, -2, ____ 1, 3, 5, 7, 9, ____ 100, 50, 25, 12.5, ____ 1000, 2000, 4000, 7000, ____ 1000, 2000, 4000, 8000, ____ 10, 10, 20, 30, 50, ____ 4, 6, 10, 16, 24, 34, ____ Enhancement Instructions: For each situation below, identify the type of sequence (arithmetic, geometric, Fibonacci, or other), determine the next term, describe the rule, and solve the question that follows. Show your complete solution and explain your reasoning. 1. Daily Step Challenge:	Activity C.4: More Exercises on Sequence These additional exercises may be given as assignments or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. <i>Answer for Remediation</i> 10, 7, 4, 1, -2, ____ Next term: -5 Rule: Subtract 3 Type: Arithmetic sequence 1, 3, 5, 7, 9, ____ Next term: 11 Rule: Add 2 Type: Arithmetic sequence 100, 50, 25, 12.5, ____ Next term: 6.25 Rule: Divide by 2 Type: Geometric sequence 1000, 2000, 4000, 7000, ____ Next term: 11,000

	A student walks 2,000 steps on Day 1 and increases their step count by 500 each day. a. How many steps will the student walk on Day 7? b. How many total steps will they have walked by the end of Day 7? 2. Investment Growth: An initial investment of ₱5,000 grows by doubling every year. a. What is the total amount after 4 years? b. How does this model help in understanding compound growth? 3. Fibonacci Pattern in Nature: The number of petals in certain flowers follows the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, ... a. What is the 10th term of the sequence? b. What makes the Fibonacci sequence unique in nature? 4. Saving for a Bike: Luis saves ₱100 in the first week and increases his savings by ₱100 each week. a. How much will Luis save in the 6th week? b. How much will he have saved in total after 6 weeks? c. If the bike costs ₱3,000, in which week will he be able to afford it? 5 Puzzle Sequence: The pattern goes: 1, 4, 9, 16, 25, ... a. What is the rule? b. What is the next term?	Rule: Add 1000, then 2000, then 3000, then 4000... Type: Pattern with increasing additions (not strictly arithmetic) 5. 1000, 2000, 4000, 8000, ____ Next term: 16,000 Rule: Multiply by 2 Type: Geometric sequence 6. 10, 10, 20, 30, 50, ____ Next term: 80 Rule: Add the two previous terms (10+10=20, 10+20=30, 20+30=50...) Type: Fibonacci-type sequence 7. 4, 6, 10, 16, 24, 34, ____ Next term: 46 Rule: Add consecutive even numbers: +2, +4, +6, +8, +10... Type: Pattern with increasing even-number additions <i>Answer for Enhancement:</i> 1. Daily Step Challenge Sequence: 2,000; 2,500; 3,000; 3,500; ... a. Day 7 step count: b. Rule: Add 500 each day Day 7: $2,000 + (6 \times 500) = 5,000$ steps Total steps by Day 7 (Sum of 7 terms): Formula: $S_n = n/2 \times (\text{first term} + \text{last term})$ $S_7 = 7/2 \times (2,000 + 5,000) = 3.5 \times 7,000 = 24,500$ steps 2. Investment Growth Sequence: 5,000; 10,000; 20,000; 40,000; 80,000 a. Amount after 4 years (5 terms): ₱80,000 b. Explanation: This models compound growth, where the value doubles consistently due to exponential increase a common principle in investments with compounding interest.
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		3. Fibonacci Pattern in Nature Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 a. 10th term: 55 b. Unique trait: Each term is the sum of the two previous terms, often found in nature (e.g., leaf arrangements, flower petals, pinecones). 4. Saving for a Bike Sequence: 100, 200, 300, 400, 500, 600, ... a. 6th week saving: ₱600 b. Total after 6 weeks (Arithmetic series): $S_6 = 6/2 \times (100 + 600) = 3 \times 700 = \text{₱}2,100$ c. To reach ₱3,000: Use: $S_n = n/2 \times [2a + (n - 1)d]$, where $a = 100$, $d = 100$ $\text{₱}3,000 = n/2 \times [2(100) + (n - 1)(100)]$ $\text{₱}3,000 = n/2 \times (200 + 100n - 100) = n/2 \times (100n + 100)$ $\text{₱}3,000 = (n/2)(100n + 100)$ Multiply both sides by 2: $\text{₱}6,000 = n(100n + 100)$ Divide both sides by 100: $60 = n(n + 1)$ Solve: $n^2 + n - 60 = 0 \rightarrow (n + 8)(n - 7) = 0$ So, $n = 7$ weeks (₱2,800), $n = 8$ weeks (₱3,600) Answer: 8 weeks 5. Puzzle Sequence Sequence: 1, 4, 9, 16, 25, ... a. Rule: n^2, where n = term position b. Next term: $6^2 = 36$
III. CONTENT	Lesson 2.2: Sequences and Series Recommended time: 2 hours	
IV. OBJECTIVES	The learners... 1. describe a series. 2. identify whether a given list of numbers is a sequence or a series. 3. explain the key difference between a sequence and a series. 4. write a given series in sigma notation and vice versa	
V. PROCEDURES	ACTIVITIES	ANNOTATIONS
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge	Activity A.1: Pattern and Term

	Activity A.1: Pattern and Term Instruction: 1. Analyze the number patterns presented on the screen. a. 2, 4, 6, ____ b. 1, 1, 2, 3, ____ c. 100, 90, 80, ____ d. Square tile patterns: 1 tile, 4 tiles, 9 tiles, ____ 2. Give the next missing term and the rule. 3. Discuss the ideas with the class, explain and give reasoning.	In this activity, learners will observe and analyze various number patterns to identify the next term and describe the rule that governs each sequence. These patterns may include arithmetic sequences, geometric sequences, the Fibonacci sequence, and square numbers. After finding the missing terms, you will explain your reasoning and discuss your ideas with the class. This activity serves as an introduction to the concept of sequences and series by helping you recognize regularities and relationships in numbers. It lays the groundwork for understanding how patterns evolve and how they can be represented and analyzed mathematically in later lessons. To facilitate it, you may apply one of the following strategies. Strategy 1: Group Work 1. You will be divided into 4 groups. Each group will work together to analyze the number patterns shown on the screen. 2. The patterns will be presented one at a time using the TV or projector. 3. Take a moment to think silently about the next term and identify the rule behind the pattern. 4. A volunteer from your group will explain your answer and reasoning to the class. 5. As a class, you will listen to each explanation and decide whether it clearly supports the correct answer. Strategy 2: Math Stations 1. You will see number patterns printed on meta cards posted in five different stations around the room. 2. You will be divided into five groups, and your group will rotate through each station.
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		3. You will have 1 minute per station to solve the number pattern posted. 4. You will ask guiding questions to help learners think deeply, such as: • Is there any pattern that you can see? • What operation is being applied? • How do you get the next number? 5. After completing all the stations, you will discuss learners' answers with the class.
	<i>A.2. Establishing the Purpose of the Lesson</i> Activity A.2: Difference and Shortcut Instructions: 1. Two sets of numbers will be presented on the board/screen. A: 3, 6, 9, 12, 15 B: $3 + 6 + 9 + 12 + 15$ 2. Observe, compare and reflect on the patterns shown. 3. Discuss using the guide questions: • What is the difference between A and B? • What do they represent? Answer: A is a list. B is a sum.	Activity A.2: Difference and Shortcut This activity is designed to help students distinguish between a sequence and a series. In this activity, you will present two related sets on the board or screen: Set A (a list of numbers forming an arithmetic sequence) and Set B (the sum of those numbers). Students will observe and compare both sets and reflect on their differences through guided questions such as, "What is the difference between A and B?" and "What do they represent?" This activity aims to develop conceptual understanding by showing how sequences build up to series, serving as a foundational step toward introducing formulas and shortcuts for calculating the sum of arithmetic sequences. To facilitate this activity, follow the given steps. Strategy: Guided Inquiry with Think-Pair-Share Instructions: 1. Present these two on the board/screen. A: 3, 6, 9, 12, 15 B: $3 + 6 + 9 + 12 + 15$ 2. Let the students individually observe, compare and reflect on the two patterns of numbers, posing the following questions: • What is the difference between A and B? • What do they represent?

		3. Students will discuss their ideas with a partner. 4. Invite pairs to share their thoughts with the class. 5. The teacher synthesizes and confirms key concepts. Guide students to realize the difference between the two patterns: Pattern A is a sequence, while Pattern B is a series. 6. Conclude by saying: “We already know about sequences. Let us now explore what happens when we add the terms in a sequence. Adding the terms in a sequence is called a series.”
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Build a Series! Instructions: Given the different sequences: 0.1, 0.2, 0.3, 0.4, -3, 0, 3, 6, 9 50, 45, 40, 35, 30 1, -2, 4, -8, 16 2, 4, 8, 16, 32 81, 27, 9, 3, 1 1. Compute the sum of the terms in each sequence. 2. Show how the terms of the sequences are added.	Activity B.1: Build a Series! To facilitate this activity the teacher may apply the strategy below. Strategy: Group Discussion Instructions: 1. Introduce the activity by saying: “We already know about sequences. Let us now explore what happens when we add the terms in a sequence.” 2. Divide the students into 5 groups. 3. Have each group add the terms of one sequence. 4. Each group will present their series to the class. 5. Discussion Prompt: What is the difference between a sequence and a series?
	B.2. Discussing the Concept Activity B.2: Lesson Discussion: Series Instructions:	Activity B.2: Lesson Discussion: Series In this session, series introduces students to the fundamental concepts related to mathematical series, building directly on their prior understanding of

	Study the information and examples carefully and prepare for class discussion. Key Terms and Definitions Series - the sum of the terms of a sequence. <i>Example:</i> For the sequence 1, 3, 5, the series is $1 + 3 + 5 = 9$ Finite Series - a series with a definite number of terms. <i>Example:</i> $2 + 4 + 6 + 8$ (only 4 terms) Infinite Series - a series that continues without end. <i>Example:</i> $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ The nth term of a sequence is usually denoted by a_n Partial Sum (S_n) - the sum of the first n terms of a sequence. If the nth term of the sequences is a_n, then $S_n = a_1 + a_2 + a_3 + \dots + a_n$ <i>Example:</i> In the sequence, 1, 3, 5, 7, 9, . . . , $S_4 = 1 + 3 + 5 + 7 = 16$ S_n is the nth term of the series Sigma Notation (Σ) - a shorthand symbol used to express a series. To write $a_1 + a_2 + a_3 + \dots + a_n$ in sigma notation: $\sum_{n=1}^n a_n$	sequences. In this activity, students will study key terms and definitions which include as series, finite and infinite series, partial sum, nth term, and sigma notation through clear examples and explanations. The purpose is to guide students in recognizing that a series is the sum of the terms of a sequence and to help them understand how to express and compute sums using both expanded and sigma notation. This activity serves as a structured input phase of the lesson, where students process essential concepts in preparation for a class discussion and deeper application of formulas in solving problems involving arithmetic and geometric series. To facilitate the discussion, teacher may select any of the provided strategies or any strategy that suit to the need of the learner. Strategy 1: Math Match-Up Instructions: 1. Prepare sets of cards: Set A: Key terms or formula names Set B: Definitions or Formula expressions Set C: Examples 2. Distribute shuffled cards to small groups. 3. Groups must match all three correctly: term, definition/formula, and example. 4. After the given time, review answers as a class. Strategy 2: Formula Hunt Instructions: 1. Use meta cards for Item 1 above. 2. Place the cards around the room or distribute them randomly. 3. Divide students into small teams. 4. Each team searches for the cards that match the term with its corresponding definition/formula. When they
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	In general, $\sum_{n=start}^{k=end} a_n \text{ (expression in terms of } a_n\text{)}$ Example: $1 + 2 + 3 + 4 + 5$ in sigma notation is $\sum_{n=1}^5 a_n$	find a correct pair, they return to the teacher for confirmation. 5. After all cards are matched, discuss each pair with the class, ensuring everyone understands how the formulas and terms connect. 6. Practice with examples using the formulas. Strategy 2: Peer Teaching Instructions: 1. Create a teaching station for each term with definition/formula. 2. At each station, prepare: The definition/formula for the term. An example problem related to the given term. 3. Split the class into small groups and assign each group to a station. 4. Each group explains the term, definition/formula, and example. 5. After the allotted time, rotate the groups to the next station. 6. When all groups have visited each station, bring the class together to discuss any challenges or solutions. Video Lesson Instructions: To reinforce student understanding of the lesson on sequences and series, you may integrate the following video material into your lesson. Title: Sequences and Series Link: https://www.youtube.com/watch?v=REDVjUTvXvk&t=368s How to Use: 1. Download or stream the video ahead of time to avoid connectivity issues in class. It provides a step-by-step
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		explanation of the concepts with examples and visuals, ideal for learners needing reinforcement. 2. Play the video during the session, pausing at key points to explain or clarify: Around 6:00 minutes, the distinction between sequences and series is clearly discussed. 3. Use the examples in the video to model how to write a series in sigma notation. 4. After the video, engage students in the Practice Problems activity to apply what they learned in a hands-on and guided way.
	B.3. Developing Mastery Activity B.3: Practice Problems- Sum It Instructions: Do the following: 1. Find the sum of the first 4 terms of the sequence, 3, 7, 11, 15, . . . Answer: $3 + 7 + 11 + 15 = 36$ 2. Find the sum of the sequence: 5, 10, 20, 40, 50 Answer: $5 + 10 + 20 + 40 + 50 = 155$ 3. Write the series, $2 + 4 + 6 + 8 + 10$ in sigma notation. Answer: The terms are $2 \times 1, 2 \times 2, 2 \times 3, 2 \times 4, 2 \times 5$. There are 5 terms in the series. In sigma notation: $\sum_{n=1}^5 2n = 30$ 4. Evaluate the series written in sigma notation:	Activity B.3: Practice Problems - Sum It In this activity, learners will firm-up their learnings by answering another set of problems. The teacher may apply any of the following strategies. Additional items may be provided. Strategy 1: Think-Pair-Solve-Share Instructions: 1. Give each student the problems in Activity 1. 2. Ask them to solve one problem alone (Think). 3. Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve). 4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). 5. Use visual aids like posters or digital slides to display each problem. Strategy 2: Color-Coded Stations Instructions: 1. Prepare 5 meta cards with one problem each (different color for each type of problem). 2. Divide class into small groups of 4. 3. Give each group one card to solve together. 4. After the allotted time, say "Pass!" and have groups pass their card clockwise to the next group.

$$\sum_{n=1}^4 (3n + 1)$$

Answer: When $n = 1$, $3(1) + 1 = 3 + 1 = 4$

When $n = 2$, $3(2) + 1 = 6 + 1 = 7$

When $n = 3$: $3(3) + 1 = 9 + 1 = 10$

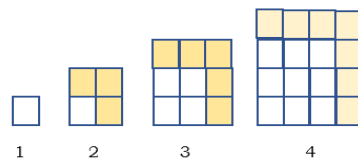
When $n = 4$: $3(4) + 1 = 12 + 1 = 13$

Then add:

$$4 + 7 + 10 + 13 = 34$$

$$\sum_{n=1}^4 (3n + 1) = 34$$

5. The figure shows a growing pattern of tiles. Count the number of tiles in each figure.



- How many tiles are there in the 5th figure?
- How many tiles are there in total from the 1st to the 7th figures?
- What is the rule used to determine the number of tiles in each figure?

- Continue until all groups have solved all cards.
- Do a quick review afterward, clarifying steps and encouraging reflection.

Strategy 3: Interactive Whiteboard Relay

Instructions:

- Divide the class into two teams.
- Display one problem at a time on the board or projector.
- A student from each team races to solve and explain their answer on the board.
- Award points for correct answers and clear explanation.
- Encourage class discussion after each round—ask if anyone solved it differently or spotted errors.

	Answer: 25 tiles 140 tiles in total - There are n^2 tiles in Figure n	
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Applications Activity C.1: Saving for a Goal Scenario: Anna decides to save money every week to buy a bicycle worth ₱3,000. She starts by saving ₱100 in the first week and increases her savings by ₱50 each week. Questions: - How much will Anna save in the 5th week? - What is the total amount saved after 5 weeks? - Will she reach her goal in 5 weeks? If not, how many weeks will it take? - How much will Anna save if she saves for 20 weeks? (You may use an Excel spreadsheet in finding the sum) Answer: Show the calculations step by step and adding terms. Let a_n be the amount saved in week n. $a_1 = 100$ $a_2 = 150$ $a_3 = 200$ $a_4 = 250$ $a_5 = 300$ Anna's savings per week is: $1^{\text{st}} \text{ week} = \text{P}100$ $2^{\text{nd}} \text{ week} = \text{P}100 + \text{P}50 = \text{P}150$ $3^{\text{rd}} \text{ week} = \text{P}150 + \text{P}50 = \text{P}200$	Activity C.1: Saving for a Goal This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output.

	4th week = P200 + P50 = P250 5th week = P250 + P50 = P300 Add her savings: P100 + P150 + P200 + P250 + P300 = P1000 Total amount saved is P 1000 $S_n = a_1 + a_2 + a_3 + a_4 + a_5$ In the 5th week she will save P 300 Anna will save P 1000 in 5 weeks No, she will not reach her goal in only 5 weeks. Anna will need 10 weeks to reach P 3000. P 11,500 if she saves for 20 weeks.	
	<i>C.2. Making Generalization</i> Activity C.2: SS in Disguise Instructions: 1. Study the following items. 2. Determine if it is a sequence or a series, if it is a series, express it in sigma notation. 3. Discuss the differences you observe between a sequence and a series as you complete the task. 2, 4, 6, 8, 10 1, 4, 9, 16, 25 2 + 4 + 6 + 8 + 10 1 + 4 + 9 + 16 + 25	Activity C.2: SS in Disguise To help students make meaningful generalizations, the teacher may use the Think-Pair-Share + Sentence Frames approach to promote reflection, discussion, and synthesis for deep understanding. Steps: 1. Think (Individual Reflection) Write your reflection on these questions: • “How can you easily tell whether a list is a sequence?” • “What role does the sigma notation play in writing a series?” • “What changes when we switch from a sequence to a series?” • “ Why is it helpful to use sigma notation when working with longer series?” 2. Pair (Collaboration) Sharing thoughts with a partner.

		Prompt: “Compare what you noticed in the lesson that is being discussed. Did you find the same rule? What general statement can you form? 3. Share (Class Synthesis) Invite at least 3 student volunteers to share their general statement with the class. Shared generalization will be written on the board. 4. Sentence Frames Use these sentence starters to help express learners generalizations: “A sequence is _____, while a series is _____. We can represent a series compactly using _____, which can help us to _____.”
	C.3. Evaluating Learning Activity C.3: Assessing Understanding on Sequence and Series Instructions: Choose the letter of the correct answer. Write your answers on a sheet of paper. 1. Which of the following best describes a <i>series</i>? a. A sum of numbers in a pattern b. A list of numbers in a specific order c. A variable expression with a pattern d. A formula for a term 2. Which of the following is a series? a. 2, 4, 6, 8 b. 3, 6, 9, 12 c. 2 + 4 + 6 + 8 d. 1, 3, 5, 7 3. What does the sigma symbol (Σ) represent in mathematics?	Activity C.3: Assessing Understanding on Sequence and Series This activity must be completed individually. Its completion is essential for assessing learners understanding of the lesson. Review their work promptly so timely intervention can be provided to those who need additional support or remediation. Additional items may be provided.

- a. A product
b. A term
c. A square root
d. A sum
4. What is the sigma notation of the series $5 + 10 + 15 + 20$?
a. $\sum_{n=1}^4 5n$
b. $\sum_{n=1}^4 5n + 1$
c. $\sum_{n=1}^4 5n - 1$
d. $\sum_{n=1}^4 5n + 5$
5. What is the sigma notation for $1 + 3 + 9 + 27$?
a. $\sum_{n=1}^4 3^n$
b. $\sum_{n=1}^4 3^{n-1}$
c. $\sum_{n=1}^4 3n$
d. $\sum_{n=1}^4 3n - 1$

C.4. Additional Activities

Activity C.4: Additional Exercises on Sequence and Series

Remediation: Sort and Represent

Instructions:

Do as indicated.

A. Sort each of the following into Sequence or Series:

Expressions	Type (Sequence/Series)
4, 8, 12, 16	
$4 + 12 + 16$	
10, 20, 30, 40	

B. Write the sigma notation for each series.

1. $4 + 8 + 16 + 32$

2. $3 + 6 + 9 + 12$

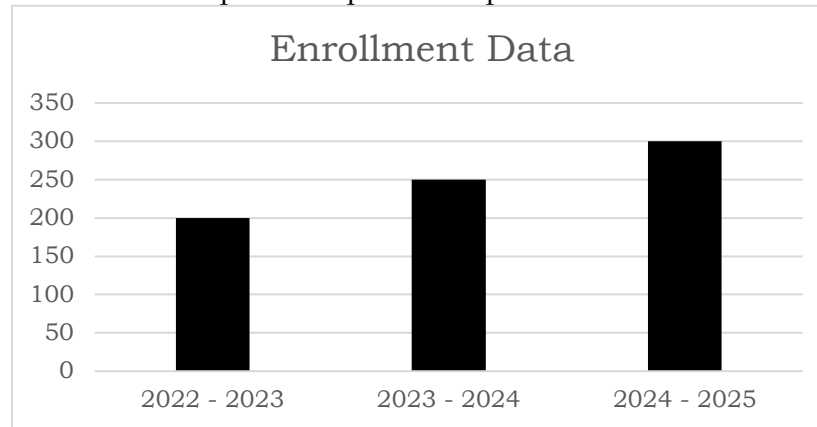
Activity C.4: Additional Exercises on Sequence and Series

These additional exercises may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

(Teachers may opt to add or give other numerical expressions)

	C. Expand the Sigma Notation and calculate the sum. 1. $\sum_{n=1}^4 5n =$ _____ 2. $\sum_{k=0}^7 (2k + 1) =$ _____ <i>Enhancement:</i> Scenario: You decided to save money weekly using a pattern. On Week 1, you save ₱50, and each week, you increase the amount saved by ₱20. <i>Tasks:</i> 1. Write the sequence for the first 8 weeks. 2. Find the total savings after 8 weeks using the formula for the sum of an arithmetic series. 3. Represent the total savings using sigma notation.	
III. CONTENT	Lesson 2.3: Arithmetic Sequence and Series Recommended time: 4 hours	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: - illustrate sequence and series. - define arithmetic sequence and series and their key components. - find the nth term of an arithmetic sequence. - use the appropriate formula $S_n = \frac{n}{2}(a_1 + a_n)$ or $S_n = \frac{n}{2}[2a + (n - 1)d]$ in calculating the sum of an arithmetic series. - solve real world problem involving arithmetic sequence and series. - demonstrate appreciation of the importance of arithmetic sequences and series by explaining or presenting how they are used in making informed decisions in real-life situations.	
IV. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	<i>A.1. Activating Prior Knowledge</i> Activity A.1: Graphical Analysis Enrollment Data Instructions: - Observe the graph of enrollment shown below. - Investigate the transformation of each bar from one school year to the next. - Identify whether the enrollment data is (a) added with the same amount each time, (b) multiplied by the same amount each time, or (c) neither.	Activity A.1: Graphical Analysis Enrollment Data This activity activates learners' prior knowledge of number patterns and prepares them for the introduction of the new concept of Arithmetic Sequences and Series. To facilitate this activity, the teacher may apply one of the following strategies: Strategy 1: Think-Pair-Share Prepare the material such as printout of the graphical presentation of the enrollment data.

4. Answer the process questions provided.



Process Questions:

- What have you observed about the trend of the graph?
- What pattern is shown in the graph? Is it increased by the same number, multiplied by the same number, or (c) neither?
- Predict the possible enrollment rate for the School Year 2025 – 2026.
- If the trend of the graph of the enrollment data continues, how many students will there be in the School Year 2029 – 2030?

- Assign a partner or study buddy who is high performing to a student who needs assistance.
- Explain to the learners that this activity serves as a foundational task that will help them connect what they already know to the current topic. Specifically, this activity supports the learning objective: "define arithmetic sequence and its key components" by guiding learners to recognize and understand the characteristics of an arithmetic sequence.
- Present the activity instructions to the learners as follows:

- Begin by solving each problem independently to form your own understanding (*Think*).
 - Next, pair up with a classmate to compare your answers and discuss your reasoning (*Pair*).
 - Finally, each pair will share their responses and thought processes with the whole class (*Share*).
- Facilitate the class discussion by briefly reviewing how each step contributes to deeper understanding and strengthens critical thinking, communication, and analytical skills essential for mastering the lesson.

Strategy 2: Technology-Enhanced Assessment

- Present the graph of the enrollment data on the board.
- Use a digital tool such as Google Forms, Kahoot, or Edmodo Quizzes to create the diagnostic test. Input the process questions as test items and provide multiple-choice options for each question.
- Explain to the learners that this activity serves as a foundational task that will help them connect what they already know to the current topic. Specifically, this activity supports the learning objective: "define arithmetic sequence and its key components" by guiding learners to recognize and understand the characteristics of an arithmetic sequence.

		4. Administer the online diagnostic test to learners individually or in groups. 5. Explain the instructions clearly: The test will contain a mix of multiple-choice or short-answer questions. There is a set time to complete the digital test. 6. Inform students that this is a low-stakes diagnostic test meant to help both them and the teacher understand their current level. 7. Facilitate the class discussion by briefly reviewing how each step contributes to deeper understanding and strengthens critical thinking, communication, and analytical skills essential for mastering the lesson. Note: The teacher may also incorporate the following real-life scenarios to further contextualize and enhance the activity. -Population Growth -Plant Growth
	A.2. Establishing the Purpose of the Lesson Activity A.2: Word Wall Match Instructions: 1. Form teams of 5 members each. 2. Each team will be given a set of word cards, definition cards, and example cards. 3. Discuss the meaning of each word card within your team. 4. Read and understand the statements on the definition cards and example cards. 5. Match each word card with its corresponding definition card and example card. 6. Post your matches on the word wall following this arrangement: Word → Definition → Example After all the groups have posted their matches, take a short walk around the room to view the answers of other groups.	Activity A.2: Word Wall Match This activity is designed to build students' foundational understanding of key concepts related to sequences and series. Working in teams, students match word cards with their corresponding definitions and real-life examples, focusing on terms such as <i>sequence</i>, <i>series</i>, <i>term</i>, and <i>Fibonacci sequence</i>. This activity encourages discussion, critical thinking, and pattern recognition as students differentiate between related ideas and organize them visually on a word wall. By engaging in this task, learners activate prior knowledge, clarify mathematical vocabulary, and prepare for a deeper exploration of arithmetic sequences and series and their applications in real-life situations. To facilitate this activity, you may apply any of the presented strategies. Strategy 1: Gallery Walk Approach Instructions:

6. Observe similarities or differences between your matches and theirs.
7. Return to your seats and be prepared to explain your group's word matches.

WORD WALL

WORD	DEFINITION	EXAMPLE
Sequence	A set of numbers arranged in a specific order or pattern	2, 4, 8, 16 1, 5, 9, 13
Series	The sum of the set of numbers according to some pattern or rule.	$2 + 6 + 10 + 14$ $6 + 9 + 12 + 15$
Term	It refers to each of the elements in a sequence or series.	2 12
Fibonacci Sequence	A sequence of numbers where each number after the first two is the sum of the two preceding one.	0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

1. Prepare 4 Word Cards, 4 Definition Cards, and 4 Example Cards. Place each set in a labeled envelope for easy distribution and organization.
2. Group learners into teams of 5 members (adjust based on class size).
3. Introduce the activity by saying:
"Let's start with a quick game to see if you already know the basic concepts of sequences and series!"
4. Distribute the envelopes containing the Word Cards, Definition Cards, and Example Cards.
5. Ask learners to read and understand the definitions and examples, then match each word card with its suitable definition and example cards.
6. After 5 minutes, have teams post their matches on the word wall in this order:
Word → Definition → Example
7. Connect to the lesson by stating:
"This activity serves as a springboard to help you grasp the foundational concepts for a deeper understanding of sequences and series."

Strategy 2: K-W-L Chart

Instructions:

1. Do this activity in groups depending on your class setup.
2. Ask the learners to complete the task by letting them:
 - discuss and write down what they already know about the terms: Sequence, Series, Terms, and Sum. (*What I Know*)
 - talk about what they're still curious about or confused by regarding these concepts. (*What I Want to Know*)
 - state what they will do to build their knowledge on the basic terms given. (*How Will I Learn*)

		• reflect on their learning after the lesson. (<i>What I Learned</i>). 3. Connect to the lesson by stating: This activity serves as a springboard to help you grasp the foundational concepts for a deeper understanding of sequences and series. Strategy 3: Anticipation-Reaction Guide This activity elicits learners’ prior knowledge of basic terms related to sequences and series. Instructions: 1. Print and provide an Anticipation-Reaction (AR) guide to each learner. 2. Instruct the students to carefully read and understand the statements in the first column of the AR Guide. In the “Before” column, ask them to write "Agree" if they agree with the statement, or "Disagree" if they do not agree. 3. At the end of the lesson, ask the learners to revisit the AR Guide and respond to the statements again, this time in the “After” column. They should reflect on how their understanding has changed and write "Agree" or "Disagree" based on their current thoughts. 4. Ask representative to share their realizations. Connect to the lesson by stating: This activity serves as a springboard to help you grasp the foundational concepts for a deeper understanding of sequences and series. Alternative Activity: Anticipation – Reaction Guide <table><tr><th>Before</th><th>Statement</th><th>After</th></tr><tr><td></td><td>Sequence is a set of numbers arranged in a specific order or pattern.</td><td></td></tr></table>	Before	Statement	After		Sequence is a set of numbers arranged in a specific order or pattern.	
Before	Statement	After						
	Sequence is a set of numbers arranged in a specific order or pattern.							

		<table> <tr> <td></td><td>Series refers to each of the element in a sequence.</td><td></td></tr> <tr> <td></td><td>A Fibonacci sequence is a sequence of numbers where each number after the first two is the sum of the two preceding ones.</td><td></td></tr> <tr> <td></td><td>$6 + 9 + 12 + 15 = 42$ is an example of sequence.</td><td></td></tr> <tr> <td></td><td>1, 5, 9, 13 is an example of sequence.</td><td></td></tr> <tr> <td></td><td>15 is one of the terms in the given number pattern, 1, 5, 9, 13.</td><td></td></tr> </table>		Series refers to each of the element in a sequence.			A Fibonacci sequence is a sequence of numbers where each number after the first two is the sum of the two preceding ones.			$6 + 9 + 12 + 15 = 42$ is an example of sequence.			1, 5, 9, 13 is an example of sequence.			15 is one of the terms in the given number pattern, 1, 5, 9, 13.	
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	15 is one of the terms in the given number pattern, 1, 5, 9, 13.																
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1.1: Discovering the Secret of the Arithmetic Sequence Task: The learners will explore a numerical pattern and derive the formula for the nth term of an arithmetic sequence. Instructions: Consider a sequence of numbers beginning with the first term a. Each term of the sequence is obtained by adding the number d. to the preceding term. Hence, the second term is $a + d$. The next (third) term is the second term plus d, or $a + d + d = a + 2d$. We list the first few terms in a table as shown below. 1. Complete the table. 2. Answer the guide questions to discover the general formula of the pattern.	In this part of the lesson, learners are able to explore arithmetic sequences and series by identifying numerical patterns, determining the common difference, and generating the next terms in a sequence. They will also examine how to find the sum of a finite number of terms in a series. Through hands-on tasks and collaborative learning, students will build a conceptual understanding of how arithmetic sequences and series work and how they can be applied to real-life situations. Facilitate the activities applying the provided strategies. Strategy: Think-Pair-Share Instructions: 1. Distribute the activity sheet Discovering the Secret of the Arithmetic Sequence 2. Ask students to silently observe the pattern and fill in the missing terms in the table.															

Table of Exploration

Term Position (n)	Term (T _n)
1	a
2	a + d
3	a + 2d
4	a + 3d
5	
6	
7	
8	
9	
n	

Guide Questions:

1. What pattern do you observe as you move from one position or term to the next?

Answer: the next term is obtained by adding d to the current term

2. What is the expression for the 5th, 6th, and 7th terms?

Answer: $T_5 = a + 4d$, $T_6 = a + 5d$, $T_7 = a + 6d$

3. Can you determine the expression for the 10th term using the pattern?

Answer: $T_{10} = a + 9d$

4. Can you write a general expression that shows the nth term?

Answer: $T_n = a + (n - 1)d$

**Activity B.1.2:
Exploring the Arithmetic Series**

3. Instruct them to reflect on the guide questions independently for 3 – 5 minutes and jot down their thoughts.

4. Let the students pair up and discuss their answers to the guide questions.

5. Encourage them to explain their reasoning to one another and compare their findings.

6. Ask each pair to try coming up with a shared formula for the nth term based on their discussion.

7. Invite a few pairs to share their thought process and formula with the class.

8. Facilitate a whole-class discussion, emphasizing how different learners identified the pattern.

9. Use a visual aid to demonstrate the increasing pattern visually (e.g. number line, manipulatives).

Strategy 2: Peer Teaching

Instructions:

1. Prepare the printed or digital copies of the activity sheet titled "Exploring the Arithmetic Series". Include space for learners to write their observations, calculations, and conclusions.

2. Ensure students have prior understanding of arithmetic sequences—particularly identifying the first term, common difference, and number of terms in a sequence.

3. Group students in pairs or small groups to promote interaction, collaboration, and peer discussion. Peer teaching works best when students have the opportunity to explain their ideas to each other.

4. Introduce the task by presenting the name Carl Friedrich Gauss, briefly sharing how he discovered a faster way to sum numbers when he was a child. This sets the stage for the learners to “rediscover” his method.

5. Distribute the activity sheet and explain that students will work together to explore the pattern and

Instructions:

1. Explore how to find the sum of an arithmetic series and derive the general formula using the strategy developed by Carl Friedrich Gauss.

2. Follow the steps carefully to discover a strategy for finding the sum of an arithmetic series.

Problem: Find the sum of the number pattern 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Step 1: Write the sum of the numbers in increasing order. Below it, write the sum in decreasing order.

Answer:

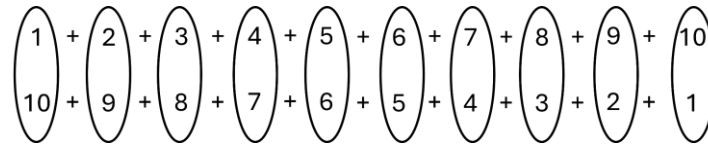
$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$$

$$10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$$

Step 2: Pair corresponding terms in the sums you wrote in

Step 3. How many pairs did you create? _____

Answer:


$$\begin{array}{cccccccccccc} 1 & + & 2 & + & 3 & + & 4 & + & 5 & + & 6 & + & 7 & + & 8 & + & 9 & + & 10 \\ 10 & + & 9 & + & 8 & + & 7 & + & 6 & + & 5 & + & 4 & + & 3 & + & 2 & + & 1 \end{array}$$

10 pairs

Step 4: What is the sum of each pair? _____

Answer: 11

Step 5: How do you get the sum of all the sums?

Answer: Since all sums are equal, we multiply by the number of sums. 10×11

Step 6: Is the sum you got in step 4 the same with the sum from 1 to 10? _____

derive a general formula for the sum of an arithmetic series.

6. Guide learners through Problem 1, where they write the sequence forward and backward, align the terms vertically, and observe how consistent sums form from pairing terms.

7. Encourage each group to complete all six steps, helping them realize how the formula $S = (n/2)(n + 1)$ emerges from their calculations.

8. Assign peer teaching roles: Have each group choose one representative to explain their group's solution and reasoning to another group. Rotate so each group explains and listens at least once.

9. Circulate around the classroom to monitor discussions, clarify misconceptions, and ask prompting questions

	Answer: No, because we added 1 to 10 twice. Step 7: Since you added the series twice, divide the result by 2. _____ Step 8: Suppose you are adding numbers from 1 to 100, do the same steps above to find the sum. Answer: We write the sum of 1 to 100 twice (increasing and decreasing order). We will get 100 pairs with each pair adding up to 101. Therefore the sum is $100 \times 101 = 10100$. Since this is twice the sum, the sum of the numbers from 1 to 100 is $10100 \div 2 = 5050$ Step 9. Provide a general formula for the sum of the numbers from 1 to n. Answer: There will be n pairs, each pair adding up to $n+1$. Therefore, the sum of the numbers from 1 to n is $n(n + 1) \div 2$.	
	B.2. Discussing the Concept Activity B.2: Lesson Discussion: Arithmetic Sequence and Series Instructions: Study the information and examples and prepare for class discussion. Key Terms and Definitions Sequence – A set of numbers arranged in a specific order or pattern. Series – The sum of the terms in sequence. Term – It refers to each of the elements in a sequence or series.	Activity B.2: Lesson Discussion: Arithmetic Sequence and Series This part of the lesson is crucial, as it serves as the foundation for learners to develop a solid understanding of arithmetic sequences and series. Through structured activities, learners will not only recognize patterns and determine the common difference but also learn to construct the general formula for the nth term of an arithmetic sequence. Furthermore, they will explore how to compute the sum of a finite arithmetic series, deepening their analytical and problem-solving skills. Mastery of these concepts is essential for applying mathematical reasoning to real-life contexts such as financial planning, budgeting, and decision-making involving repeated or accumulated values. This phase

	Arithmetic Sequence – a special type of number pattern in which every term after the first is obtained by adding a number to the preceding term. This means that in an arithmetic sequence the difference between two consecutive terms is constant. First term (a_1) – the very first value in the sequence. It is the starting point from which other terms in the sequence are produced. nth term – it refers to the general or specific term at the position n in the sequence. Common Difference – it is the constant difference between the two consecutive terms in an arithmetic sequence. It is also the number added to a term to get the next term. Arithmetic Series – it is the sum of terms of an arithmetic sequence. It is denoted by S_n. S_n is the sum of the first nth terms of a sequence. $S_n = a_1 + a_2 + \dots + a_n$ Arithmetic Sequence Arithmetic Sequence is a special type of number pattern in which the two consecutive terms have a constant difference called the common difference (d). To Calculate the nth term of an arithmetic sequence, use the formula:	ensures that learners build both procedural fluency and conceptual understanding on arithmetic sequence and series. Choose any of the suggested strategies or any approach that suited to your learners to facilitate this activity. Additional examples may be provided as needed. Strategy 1: Interactive Discussion Instructions: 1. Prepare a visual aid or PowerPoint presentation. 2. Print and provide the learners with the examples in solving the Arithmetic Sequence and Series. 3. After presenting the key terms and definitions, ask the learners to share their understanding based on the collaborative discussion they had during peer teaching. 4. From the notes on solving the arithmetic sequence 5. From the notes on solving arithmetic sequences and series, select representatives to discuss the process of calculating the nth term of an arithmetic sequence and using the appropriate formula to find the sum of an arithmetic series. 5. Afterwards, facilitate the discussion and provide feedback. Strategy 2: Jigsaw Puzzle Instructions: 1. Group the learners into 5 working teams. 2. Assign each team an example involving the calculation of an arithmetic sequence or solving an arithmetic series. 3. Afterward, regroup the learners so that each member can share what they learned from their original group. 4. Facilitate the discussion and provide feedback.
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$$a_n = a_1 + (n - 1)d$$

where:

a_n = the n th term

a_1 = the first term of the arithmetic sequence

d = the common difference

n = the position of the term in the sequence

Example 1: Find the 10th term of the arithmetic sequence:
5, 8, 11, 14, ...

Solution:

Step 1: Identify the value of the variables of the formula.

$$a_1 = 5$$

$$d = 3$$

$$n = 10$$

$$a_n = ?$$

Step 2:

Substitute the values to the formula. Then, solve for a_n .

$$a_{10} = 5 + (10 - 1)3$$

$$= 5 + (9)3$$

$$= 5 + 27$$

$$a_{10} = 32$$

Therefore, the 10th term in the arithmetic sequence is 32.

Example 2: In the arithmetic sequence: 3, 9, 15, 21, ... ;
find the value of n if $a_n = 75$.

Solution:

Step 1: Identify the value of the variables of the formula.

$$a_1 = 3$$

$$d = 6$$

$$a_n = 75$$

$$n = ?$$

Substitute the values to the formula. Then, solve for n .

$$75 = 3 + (n - 1)6$$

$$75 = 3 + 6n - 6$$

$$75 - 3 + 6 = 6n$$

$$78 = 6n$$

$$n = 13$$

Therefore, 75 is in the 13th term of the arithmetic sequence.

Example 3: The 5th term of an arithmetic sequence is 20, and the 12th term is 48. Find the following: (a) common difference, (b) first term, and (c) the 30th term.

Solution:

Step 1: Identify the given values.

$$a_5 = 20$$

$$a_{12} = 48$$

Step 2: Substitute the given values to the formula

$$\text{For } a_5 = 20$$

$$20 = a_1 + (5 - 1) d$$

$$20 = a_1 + 4d \text{ (Equation 1)}$$

$$\text{For } a_{12} = 48$$

$$48 = a_1 + (12 - 1) d$$

$$48 = a_1 + 11d \text{ (Equation 2)}$$

Step 3: Subtract the equations

$$20 = a_1 + 4d$$

$$48 = a_1 + 11d$$

$$-28 = -7d$$

$$d = 4$$

Step 4: Substitute the value of $d = 4$ to equation 1:

$$20 = a_1 + 4(4)$$

$$20 = a_1 + 16$$

$$20 - 16 = a_1$$

$$a_1 = 4$$

Step 5. Solve for the 30th term.

$$a_{30} = 4 + (30 - 1) 4$$

$$= 4 + (29)4$$

$$= 4 + 116$$

$$a_{30} = 120$$

Therefore, the common difference is 4, first term is 4, and the 30th term is 120.

Example 4: During the dry season, a farmer in your town is testing a new water irrigation system for his rice field. After 1 minute of turning on the water pump, the water level in the irrigation canal rose to 30 centimeters. After 2 minutes, it reached 85 centimeters, and after 3 minutes, the level was 140 centimeters. If the water level continues to rise at this constant rate, how high will the water level be after 16 minutes?

Solution:

Let a_n = the water level after n minutes

Step 1: Identify the given.

$$a_1 = 30\text{cm}$$

$$d = 55 \text{ cm}$$

$$n = 16 \text{ min}$$

$$a_n = ?$$

Step 2:

Substitute the given values to the formula.

$$a_{16} = 30 + (16 - 1)55$$

$$= 30 + (15) \cdot 55$$

$$= 30 + 825$$

$$a_{16} = 855 \text{ cm}$$

Therefore, after 16 minutes, the water level will be 855 centimeters high in the irrigation canal.

Arithmetic Series

Arithmetic Series is the sum of terms of an arithmetic sequence. It represents adding the terms of the sequence together.

Let's say we have the arithmetic sequence: 3, 6, 9, 12. This is an arithmetic sequence where the common difference $d = 3$. Now, let's add the terms: $3 + 6 + 9 + 12 = 30$. The sum is an example of arithmetic series.

Let us compute the sum of the first n terms of an arithmetic series.

$$S_n = a_1 + a_2 + \dots + a_n$$

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \dots + [a_1 + (n - 1)d]$$

There are n terms, so we add together all the a_1 's

$$S_n = na_1 + [1 + 2 + \dots + (n - 1)]d$$

We have already derived the formula for the sum of the numbers from 1 to $n-1$.

$$S_n = na_1 + (n - 1)nd/2 = n[2a_1 + (n - 1)d]/2$$

Now we know that $a_1 + (n - 1)d = a_n$. Therefore,
 $S_n = n(a_1 + a_n)/2$

We therefore have the following formulas to compute the sum of the first n terms of an arithmetic series:

If the last n th term (a_n) is given: $S_n = \frac{n}{2}(a_1 + a_n)$, or

If the last n th term (a_n) is not given:

$$S_n = \frac{n}{2}[2a_1 + (n - 1)d]$$

where:

S_n = the sum of first n terms

a_1 = first term

a_n = last n th term

d = common difference

Example 1: Find the sum of the arithmetic sequence with 10 terms: 3, 10, 17, ..., 66.

Solution:

Since the last n th term is given, use the 1st formula.

Step 1: Identify the given values.

$$a_1 = 3$$

$$a_n = 66$$

$$n = 10$$

$$S_n = ?$$

Step 2: Substitute the values to the formula.

$$\begin{aligned} S_{10} &= \frac{10}{2} (3 + 66) \\ &= 5 (69) \\ S_{10} &= 345 \end{aligned}$$

Therefore, the sum of the arithmetic sequence is 345.

Example 2: Find the sum of the first 15 terms of the arithmetic sequence: 4, 9, 14, 19, ...

Solution:

Since the last n th term is not given, use the 2nd formula.

Step 1: Identify the given values.

$$a_1 = 4$$

$$n = 15$$

$$d = 5$$

$$S_n = ?$$

Step 2: Substitute the given values to the formula.

$$\begin{aligned} S_{12} &= \frac{15}{2} [2(4) + (15 - 1)5] \\ &= 7.5 [8 + (14)5] \\ &= 7.5 [8 + 70] \\ &= 7.5 [78] \\ S_{12} &= 585 \end{aligned}$$

Therefore, the sum of the arithmetic sequence is 585.

Example 3: Mang Tonyo is a kalesa driver in the historic streets of Vigan, Ilocos Sur. On his first day, he earns ₱200. As more tourists visit throughout the week, his earnings increase by ₱100 each day. If this pattern continues for 15 days, how much will Mang Tonyo earn in total?

Solution:

Since the last earning or nth term is not given, use the 2nd formula.

Let a_n = the earning of Mang Tonyo on day n.

Step 1: Identify the given values.

$$a_1 = \text{₱}200$$

$$d = \text{₱}100$$

$$n = 15 \text{ days}$$

$$S_n = ?$$

Step 2: Substitute the given values to the formula.

$$S_{15} = \frac{15}{2} [2(200) + (15 - 1)100]$$

	$= 7.5 [400 + (14)100]$ $= 7.5 [400 + 1400]$ $= 7.5 [1800]$ $S_{15} = \text{₱}13,500$ Therefore, Mang Tonyo will earn a total of ₱13,500 in 15 days.	
	B.3. Developing Mastery Activity B.3: Practice Problems Instructions: Solve the following. - Find the 8th term of the arithmetic sequence: 5, 11, 17, 23, ... - The 5th term of an arithmetic sequence is 40, and the common difference is 6. Find the 1st term of the sequence. - Find the sum of the first 12 terms of the arithmetic sequence: 4, 7, 10, ... - What is the sum of the first 15 terms of the sequence: 3, 6, 9, ...? - Mang Ernesto, a jeepney driver in Manila, starts with a fare of ₱20 on his first trip of the day. For each succeeding trip, he earns ₱5 more than the previous trip. How much will he earn in total after 10 trips?	Activity B.3: Practice Problems In this activity learners will firm-up their learnings by answering another set of problems. Facilitate the activity with this strategy. Additional items may be provided as needed. Instructions: - Give each student a copy of the practice problems. - Ask them to solve one problem alone (Think). - Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve). - Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). - You may use visual aids like posters or digital slides to show each problem.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Role Playing Saving for a Family Trip Context: Imagine you and your family are planning a trip to Tagaytay. To save for the trip, you plan to save a certain	Activity C.1: Role Playing Saving for a Family Trip This activity is intended for group work, where all learners are expected to actively participate and collaborate. It provides an opportunity to deepen their understanding and enhance their skills in preparation for the upcoming performance task. Encourage learners

	amount each week, starting with ₱200 on the first week and increasing the amount by ₱50 each week. Task: Determine how much money you will save in the 8th week and how much in total your savings will be after that time. Instructions: 1. Understand the Problem: • Determine Key Information • First term a_1 • Common difference d • Number of terms 2. Find the amount saved in the 8th week (the last week). 3. Find the Total Savings. 4. Answer the Questions: • How much will you save in the 8th week? • How much in total will you have saved after 8 weeks? Reflection: • Discuss how you can apply this concept to other savings plans, like saving for a school project or a new gadget. • What could be the benefits of using an arithmetic sequence in planning savings?	to produce thorough and high-quality work. Apply Experiential-Collaborative Learning by facilitating meaningful interactions and providing constructive feedback on their output.
	<i>C.2. Making Generalization</i> Activity C.2.1: 3 – 2 – 1 Ticket Instructions: Complete the ticket given.	For the summary, the most appropriate approach is Reflective Learning. To facilitate this, you may choose and implement one of the suggested activities.

Three things I learned today:

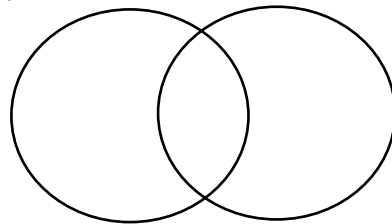
Two things I found interesting:

One question I still have in mind:

Activity C.2.2
Comparing Arithmetic Sequence and Series using a Venn Diagram

Instructions:

1. Draw a Venn Diagram. Label the left side with “Arithmetic Sequence” and the right side with “Arithmetic Series”.
2. Fill in the Venn Diagram. On the left side write ideas related to Arithmetic Sequence, on the right side write ideas on the Arithmetic Series, and the middle section are shared ideas.



C.3. Evaluating Learning

Activity C.3:

Activity C.3:
Assessing Knowledge and Skills on Arithmetic Sequence and Series

**Assessing Knowledge and Skills on Arithmetic
Sequence and Series**

Part I. Multiple Choice. Choose the best answer.

1. What is the main difference between a sequence and a series?

- A. A sequence has a formula, while a series does not
- B. **A sequence is an ordered group of terms, while a series is their sum**
- C. A sequence includes variables, while a series includes constants
- D. A sequence is always infinite, while a series is finite

2. Which of the following best defines a sequence?

- A. The difference between two terms
- B. A list of numbers without any order
- C. **A set of numbers arranged in a specific order or pattern**
- D. The sum of terms in an equation

3. What does the term “series” refer to in mathematics?

- A. A pattern that repeats
- B. A group of equations
- C. **The sum of the elements in a sequence**
- D. The position of an element in a list

4. Which formula is used to find the n th term of an arithmetic sequence?

$$a_n = a_1 + (n - 1)d$$

$$a_n = a_1 + (n + 1)d$$

$$S_n = \frac{n}{2}(a_1 + a_n)$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

5. What is the common difference of the arithmetic sequence: 6, 10, 14, 18, ...?

- A. 2 B. **4** C. 5 D. 3

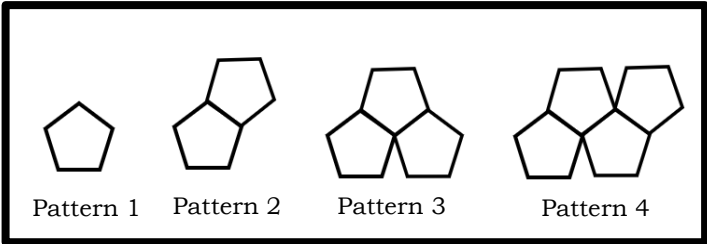
6. What is the sum of the first 8 terms of the arithmetic series: 4, 8, 12, 16, ...?

- A. 120 B. 136 C. **144** D. 160

Each learner must complete this activity individually, as it serves as a key tool for assessing their understanding of the lesson. Review their responses promptly to ensure timely and appropriate interventions for those who may need additional support or remediation.

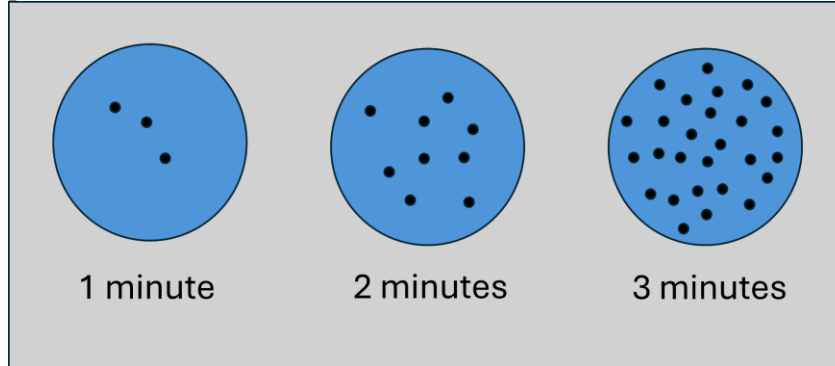
	7. In an arithmetic sequence, the 5th term is 12 and the 12th term is 40. What is the common difference (d)? A. 2 B. 3 C. 4 D. 5 8. Which of the following represents the sum of the first n terms of an arithmetic sequence? A. $S_n = a_1 + (n - 1)d$ B. $S_n = \frac{n}{2} (2a_1 + (n - 1)d)$ C. $S_n = \frac{n}{2} (a_1 + a_n)$ D. Both B and C <i>Part II. Solve the problem. Show your solution.</i> Maria is a student who wants to buy a new bike. She saves ₱50 on the first day, and every day after that, she saves an additional ₱20. She continues to save for 15 days. Questions: 1. How much money will Maria save on the 15th day? 2. What is the total amount saved after the 15 days? <i>Part III. Reflective Essay. Write a reflective essay of at least 200 words discussing the following:</i> How do you plan to use the knowledge of arithmetic sequences and series in future lessons or in your daily life?	
	C.4. Additional Activities Activity C.4: Improving/Enhancing Understanding in Arithmetic Sequence Remediation Instructions: 1. Group Formation: Divide the class into small groups of 3–4 members. Ensure a mix of skill levels in each group so that struggling students receive peer support.	Activity C.4: Improving/Enhancing Understanding in Arithmetic Sequence This supplementary activity may be assigned as homework. For in-class support, small group remediation should be provided to learners who have not yet mastered the lesson. Each group should consist of a mix of high-performing, average, and low-performing learners to encourage peer support and collaborative learning. Meanwhile, enrichment activities should be

	2. Warm-Up: Pattern Recognition: Each group receives a short set of number patterns and answers the question: "Is this an arithmetic sequence? If yes, what is the common difference?" Example: 5, 8, 11, 14, ____ 2. 20, 16, 12, 8, ____ 3. Group Task: "Complete the Sequence". Each group is given 2–3 arithmetic sequences with missing terms or values. Tasks include: • Identifying the first term (a_1) • Finding the common difference (d) • Writing the general formula [$a_n = a_1 + (n - 1) d$] • Computing the 10th or 15th term as required Example task: • The 3rd term of a sequence is 10 and the 6th term is 19. Find the first term and common difference. • Find the 12th term using the formula. 4. Peer Teaching and Feedback: Groups take turns presenting one of their problems and solutions. Other groups can ask questions or suggest improvements. Enhancement Instructions: Solve the following. 1. In a classroom, the first row has 5 desks, the second row has 7 desks, the third row has 9 desks, and so on, increasing by 2 desks per row. If there are 10 rows in total, how many desks are there altogether?	offered to those who are ready for more advanced tasks to further deepen their understanding.
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	2. A school building has stairs with decreasing steps on each level. The top level has 10 steps, the second has 12 steps, the third has 14 steps, and so on. If there are 8 levels, how many steps are there in total from the ground floor to the top? 3. Percy is making patterns with toothpicks as follows:  Pattern 1 Pattern 2 Pattern 3 Pattern 4 • How many toothpicks will Percy need for Pattern 15? • How many toothpicks will Percy have used after doing Patterns 1 to 15?	
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III. CONTENT	Geometric Sequence and Series	
IV. OBJECTIVES	At the end of the lesson, the learners shall be able to: 1. illustrate the geometric sequence. 2. define geometric sequences and series, including their key components. 3. calculate the nth term of a geometric sequence. 4. compute the geometric series. 5. solve problem involving geometric sequences and series. 6. demonstrate appreciation of the importance of arithmetic sequences and series by explaining or presenting how they are used in making informed decisions in real-life situations.	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	<i>A.1. Activating Prior Knowledge</i> Activity A.1: Situation Analysis Bacterial Growth Instructions: 1. Observe the three petri dishes containing bacteria. 2. Analyze the changes in the number of bacteria in each petri dish over time.	Activity A.1: Situation Analysis Bacterial Growth This activity activates learners' prior knowledge of number patterns and prepares them for the introduction of the new concept of Geometric Sequences and Series. To facilitate this activity, the teacher may apply one of the following strategies:

3. Determine whether the bacterial population exhibits a pattern of multiplication or not.
4. Respond thoughtfully to the process questions provided.



Process Questions:

- How will you describe the three petri dishes shown above?
- What pattern is shown in the graph?
- What would be the expected bacterial population in 4 minutes? 8 minutes?

Strategy 1: Think-Pair-Share

1. Provide the following materials if necessary:
 - A printout of the bacterial population data in the petri dishes.
2. Assign a high-performing partner or study buddy to students who need assistance.
3. Explain to the learners that this activity serves as a foundational task that will help them connect what they already know to the current topic.
4. Present the activity instructions to the learners as follows:
 - a. Begin by solving each problem independently to form your own understanding (*Think*).
 - b. Next, pair up with your assigned study buddy to compare your answers and discuss your reasoning (*Pair*).
 - c. Finally, each pair will share their responses and thought processes with the whole class (*Share*).
5. Facilitate the class discussion by briefly reviewing how each step contributes to deeper understanding and strengthens critical thinking, communication, and analytical skills essential for mastering the lesson.

A.2. Establishing the Purpose of the Lesson

Activity 2. Minute to Win It What's my pattern?

Instructions:

1. You will be divided into working teams of 5 members.
2. Your group will receive a mystery number pattern written on a meta card. You will determine the pattern rule.

The Mystery Number Patterns

3, 6, 12, _____, _____

5 + 20 + 80 + _____ + _____

Activity 2. Minute to Win It What's my pattern?

This activity is designed to help learners discover patterns in geometric sequences and series. By identifying rules in number patterns, students begin to understand how geometric sequences work, preparing them for the formal lesson and deeper discussions.

To facilitate this activity, you may apply any of the following strategies:

Strategy 1: Cooperative Group Work

Instructions:

	10, 5, $\frac{5}{2}$, $\frac{5}{4}$, _____ 1, 2, 4, 8, _____, _____, _____ 3. After one minute, groups will take turns rotating. You will move to the next mystery number pattern. Continue this until you have solved all the mystery number patterns. 4. After all mystery number patterns have been guessed, you will be given meta strips to write your answers and show them to the class. 5. Select a representative from your group to share brief insights. “How did you arrive at the missing number in the given number patterns?”	1. Organize the class into groups of five and prepare meta cards with the given mystery number patterns. 2. Each group will analyze a pattern for one minute before rotating to the next. This promotes peer discussion, quick critical thinking, and collaborative problem-solving. 3. After all patterns are visited, provide meta strips for each group to finalize and present their answers. 4. Facilitate a brief sharing session where a group representative explains their reasoning. 5. Emphasize the mathematical thinking behind identifying the structure of geometric sequences. 6. Encourage learners to verbalize their pattern recognition process to deepen understanding. Strategy 2: Gallery Walk with Station-Based Tasks Instructions: 1. Instead of rotating the students, set up a <i>Gallery Walk</i> with numbered stations placed around the room, each containing a large sheet or poster showing one mystery number pattern. 2. Assign groups to begin at different stations. Each group will discuss and solve the pattern at their assigned station, then write their answer and brief explanation directly on the station sheet. 3. After a set time (e.g., 2–3 minutes), groups rotate clockwise to the next station, review previous answers, and either confirm, add, or challenge them. 4. Once all stations have been visited, conduct a class-wide debrief where groups revisit their original station and summarize the most common or accurate reasoning presented. 5. This approach allows for deeper inquiry, visible learning, peer critique, and promotes movement, discussion, and reflection throughout the activity.
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B. Instituting New Knowledge

B.1. Presenting Examples

Activity B.1: The Secret of the Geometric Sequence

Instructions:

Consider a sequence of numbers beginning with the first term a . Each term of the sequence is obtained by multiplying the preceding number by r . Hence, the second term is ar . The next (third) term is the second term times r , or $ar \times r = ar^2$. The first few terms in a table as shown below.

Tasks:

1. Complete the table.
2. Answer the guide questions to discover the general formula of the pattern.

Table of Exploration

Term Position (n)	Term (T_n)
1	a
2	ar
3	ar^2
4	ar^3
5	
6	
7	
8	
9	
n	

Guide Questions:

- What pattern do you observe as you move from one position or term to the next?
- What is the expression for the 5th, 6th, and 7th terms?
- Can you determine the expression for the 10th term using the pattern?

Activity B.1: The Secret of the Geometric Sequence

This activity aims to guide learners in discovering the general formula for the n th term of a geometric sequence. By completing the table and analyzing how each term changes, students will recognize the consistent pattern of multiplication and develop a deeper understanding of how geometric sequences differ from arithmetic sequences.

To facilitate this activity, perform one of the provided strategies.

Strategy 1: Guided Inquiry-based Approach

Instructions:

1. Prepare the printed or digital copies of the activity sheet, visual aid, and manipulatives.
2. Ensure students have prior understanding of sequences and have encountered arithmetic sequences before introducing geometric ones.
3. Organize students in pairs or small groups to encourage discussion.
4. Start with a quick review. Introduce a relatable context or scenario (e.g., doubling bacteria, stacking cups).
5. Distribute the activity sheet.
6. Present the pattern: $a, ar, ar^2, ar^3, \dots$
7. Explain the table and guide students to fill in missing terms.
8. Encourage collaboration: students can work in pairs or small groups.
9. Walk around the room to provide hints and ask guiding questions.
10. Invite students to share their patterns and formulas.
11. Compare this formula to the arithmetic formula to reinforce the difference.

	• Can you write a general expression that shows the nth term? • How does this pattern differ from an arithmetic sequence?	Strategy 2: Learning by Doing Instructions: 1. Prepare printed or digital copies of the activity sheet, visual aids (e.g., images or diagrams of geometric growth), and physical manipulatives (e.g., paper cups, counters, or tiles to model multiplying patterns). 2. Ensure students have prior understanding of sequences, particularly arithmetic sequences, before introducing geometric sequences. 3. Organize students in pairs or small groups to support peer discussion and collaborative learning. 4. Start with a quick review of arithmetic sequences. Then introduce a concrete, relatable context—such as a paper cup tower where each level doubles, or a scenario involving rapid population growth. 5. Distribute the activity sheet to each group. 6. Present the geometric sequence pattern explicitly: a, ar, ar^2, ar^3, ... 7. Explain the table of exploration and instruct students to observe the changes between terms and fill in the missing values. 8. Encourage group discussion and collaborative problem-solving as students explore the pattern and answer the guide questions. 9. Walk around the classroom to monitor engagement, provide prompts when needed, and ask guiding questions.
	B.2. Discussing the Concept Activity B.2: Understanding the Geometric Sequence and Series Instructions: Perform the different tasks to be given by your teacher and participate actively in the discussion.	Activity B.2: Understanding the Geometric Sequence and Series In this session, students will deepen their understanding of geometric sequences and series by building on their prior knowledge of sequences. The activity introduces essential terms and concepts such as geometric series, finite and infinite series, partial sum, nth term, and sigma notation. Through guided examples and explanations, students will learn that a series

	Geometric Sequence is a special type of sequence where each term after the first is obtained by multiplying the preceding term by a fixed non-zero number. The fixed multiplier is called the common ratio, denoted by r. It is called the common ratio because it is the ratio of any two consecutive terms in the sequence. $r = a_{n+1}/a_n$ The formula in calculating the nth term in a geometric sequence is $a_n = a_1 r^{n-1}$ where: a_n = the last nth term a_1 = first term r = common ratio n = the position of the term To calculate the nth term of a geometric sequence, let's follow the steps: 1. Identify the value of the first term. 2. Find the value of the common ratio. 3. Apply the formula in calculating the nth term. Example 1: Find the 5th term of a geometric sequence where the first term is 5 and the common ratio is 3. Solution: $\begin{aligned} a_1 &= 5 \\ r &= 3 \\ n &= 5 \\ a_n &= ? \end{aligned}$ $\begin{aligned} a_5 &= 5(3)^{5-1} \\ &= 5(3)^4 \\ &= 5(81) \end{aligned}$	represents the sum of terms in a sequence and how to express and calculate this sum using both expanded and sigma notation. This activity serves as the structured input phase of the lesson, preparing students for class discussions and the application of formulas in solving real-life problems involving geometric sequences and series. The teacher may use any suitable strategy or approach that fits the learners' needs to facilitate effective learning. Additional examples may be given to the learners. Strategy 1: Explicit Instruction Instruction: 1. Use direct instruction to explicitly explain the concept of geometric sequences and series. 2. Begin by defining key terms (first term, common ratio, nth term) and introduce the standard formulas. 3. Guide learners through each example step-by-step, making sure to emphasize how values are substituted and computed. Integrate relatable real-life contexts—such as stacking cups, counting steps, and saving money—to illustrate how geometric sequences naturally occur in everyday life. 3. Encourage class discussion after each example and check for understanding through quick oral or written reflections. 4. Reinforce learning by allowing students to solve parallel problems using the same process, gradually moving from teacher-led examples to independent work. Strategy 2: Math Quest Game-Based Learning Instruction: 1. Transform the lesson into an interactive "Math Quest" game where learners work in small groups to
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$$a_5 = 405$$

Therefore, 405 is the value of the 5th term in the sequence.

Example 2: Find the 10th term of the geometric sequence:
4, 12, 36, ...

Solution:

$$a_1 = 4$$

$$r = 3$$

$$n = 10$$

$$a_n = ?$$

$$\begin{aligned} a_{10} &= 4(3)^{10-1} \\ &= 4(3)^9 \\ &= 4(19683) \\ a_{10} &= 78732 \end{aligned}$$

Example 3: The 4th term of a geometric sequence is 64, and the 9th term is 486. Find (a) the common ratio, (b) the first term, and (c) the 7th term.

Solution:

From a_4 :

$$64 = a_1 r^{4-1}$$

$$16 = a_1 r^3 \text{ (Equation 1)}$$

From a_9 :

$$486 = a_1 r^{9-1}$$

$$486 = a_1 r^8 \text{ (Equation 2)}$$

Divide the equation 2 by the equation 1:

Solving for the common ratio (r).

$$\frac{486}{64} = \frac{a_1 r^8}{a_1 r^3}$$

$$\frac{243}{32} = r^5$$

solve real-life problem scenarios involving geometric sequences and series.

2. Create activity stations or digital slides (if tech is available), each containing a challenge related to the lesson (e.g., finding the nth term, determining the sum of a series, or solving for missing values).

3. Assign roles within each group (e.g., Calculator, Recorder, Speaker, Checker) to ensure participation.

4. Provide each team with a “quest scroll” that outlines the storyline (e.g., “Help Mang Tonyo save enough money for his store expansion!”).

5. Groups earn points or “badges” for each correctly completed task. Debrief by letting each group explain their solutions and strategies.

6. This promotes peer teaching, engagement, and conceptual understanding in a fun and meaningful context.

$$\frac{3}{2} = r$$

Substitute $r = 3/2$ in equation 2:
Solving for the first term (a_1)

$$64 = a_1 \left(\frac{3}{2}\right)^3$$
$$\frac{512}{27} = a_1$$

Solve for the 7th term:

Solution:

$$a_1 = \frac{512}{27}$$

$$r = \frac{3}{2}$$

$$n = 7$$

$$a_n = ?$$

$$a_7 = \frac{512}{27} \left(\frac{3}{2}\right)^{7-1}$$

$$= \frac{512}{27} \left(\frac{3}{2}\right)^6$$

$$= 216$$

Example 4: A local farmer named Mang Pedro started raising native chickens. In his first month, he had 3 chickens. Every month, the number of chickens tripled as each chicken hatched more chicks. The number of chickens each month follows a geometric sequence.

Problem:

1. How many chickens does Mang Pedro have by the 5th month?
2. How many chickens does he have by the 8th month?

3. Write the general formula to find the number of chickens in the nth month.

Solutions:

Given:

First term $a_1 = 3$

Common ratio $r = 3$

1. 5th month

$$a_5 = 3 (3)^{5-1}$$

$$= 3 (3)^4$$

$$= 3 (81)$$

$$a_5 = 243 \text{ chickens}$$

2. 8th month

$$a_8 = 3 (3)^{8-1}$$

$$= 3 (3)^7$$

$$= 3 (2187)$$

$$a_8 = 6561 \text{ chickens}$$

3. General Formula

$$a_n = 3 (3)^{n-1}$$

$$a_n = 3^n$$

Geometric Series

Review of terms:

1. First Term (a_1): This is the very first term of the sequence. It's the starting value from which all other terms are derived by multiplication with the common ratio.

2. Common Ratio (r): This is the constant factor by which each term in the sequence is multiplied to get the next term. It is the ratio of all consecutive terms in the series.

4. S_n : This is the sum of the first n terms of a geometric sequence.

To derive the formula for the sum of the first n terms of a geometric sequence.

A geometric sequence has:

First term: a_1

Common ratio: r

Number of terms: n

When $r \neq 1$, let:

$$S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \quad \text{eq. (1)}$$

Now, multiply both sides of equation (1) by r :

$$rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \quad \text{eq. (2)}$$

Next, subtract equation (2) from equation (1):

$$\begin{array}{r} S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \\ - \\ rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \end{array}$$

$$S_n - rS_n = a_1 - a_1r^n$$

Extract S_n as a common factor on the left-hand side of the equation and a_1 on the right-hand side.

$$S_n(1 - r) = a_1(1 - r^n)$$

Now, divide both sides $1 - r$.

$$S_n = \frac{a_1(1 - r^n)}{(1 - r)}$$

This gives us a formula for a finite geometric series with n terms.

What happens if the series is infinite? This means that n becomes very large. If we look at the expression on the right side of the equation, the part affected by n is r^n . If r is "large" ($r > 1$ or $r < -1$), r^n also becomes large and we don't get a real number value for S_n . If r is "small" ($-1 < r < 1$), then r^n becomes very small for large values of n . We can therefore say that as the number of terms go to infinity ($n \rightarrow \infty$), the values of r^n become very small ($r^n \rightarrow 0$). Which means that the numerator in the expression for S_n will go to $a_1(1 - 0) = a_1$.

Therefore, if $-1 < r < 1$, the sum of an infinite geometric series is

$$S = \frac{a_1}{1 - r}$$

Example 1: Find the sum of the first five terms of the geometric series: $2 + 6 + 18 + 54 + \dots$

Solution:

$$a_1 = 2$$

$$r = 3$$

$$n = 5$$

$$S_n = ?$$

$$S_5 = 2 \left(\frac{1 - 3^5}{1 - 3} \right)$$

$$S_5 = 2 \left(\frac{1 - 243}{1 - 3} \right)$$

$$S_5 = 2 \left(\frac{-242}{-2} \right)$$

$$S_5 = 2 (121)$$

$$S_5 = 242$$

Example 2: Consider the geometric series: $10 + 5 + 2.5 + 1.25 + \dots$. Find the sum of the infinite series.

Solution:

$$a_1 = 10$$

$$r = \frac{1}{2}$$

$$S = ?$$

$$S_n = \frac{10}{1 - \frac{1}{2}}$$

	$S_n = \frac{10}{\frac{1}{2}}$ $S_n = 10 \times 2$ $S_n = 20$ Example 3: Mang Tonyo owns a small sari-sari store in the barangay. One day, he decided to start saving coins in an alkansya (piggy bank). On the first day, he puts ₱20. Each following day, he doubles the amount he saves from the previous day. He plans to continue this for 7 days. How much total money will Mang Tonyo have saved in the alkansya after 7 days? Solution: Let a_n = The money Mang Tonyo will save in the alkansya on the nth day. $a_1 = 20$ $r = 2$ $n = 7$ Use the Geometric Series Formula: $S_7 = a_1 \frac{1 - r^n}{1 - r}$ $S_7 = 20 \left(\frac{1 - 2^7}{1 - 2} \right)$ $S_7 = 20 \left(\frac{1 - 128}{1 - 2} \right)$ $S_7 = 20 \left(\frac{-127}{-1} \right)$ $S_7 = 20(127)$ $S_7 = ₱2540$ Therefore, Mang Tonyo will save ₱2540 in 7 days. B.3. Developing Mastery Activity B.3:	The teacher can guide student exploration on what happens to r^n as $n \rightarrow \infty$. For example, get the students to calculate $2^3, 2^{10}, 2^{20}, 2^{50}, \dots$ $(-1.3)^5, (-1.3)^{10}, (-1.3)^{20}, (-1.3)^{50}, \dots$ $(0.4)^3, (0.4)^{10}, (0.4)^{20}, (0.4)^{50}, \dots$ etc. Activity B.3: Practicing the Skills on Geometric Sequence and Series
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	Practicing the Skills on Geometric Sequence and Series Instructions: Solve the following. 1. Given the geometric sequence: 3, 6, 12, 24, ... Find the 7th term of the sequence. Answer: 192 2. In a geometric sequence, the first term is 5 and the common ratio is 3. Find the value of the 10th term. Answer: 98415 3. The 4th term of a geometric sequence is 16, and the 7th term is 128. Find the: - Common ratio - First term 9th term Answer: 2 2 512 4. Find the sum of the first 6 terms of the geometric series: $2 + 4 + 8 + 16 + \dots$ Answer: 126 5. A bamboo tree grows such that each week its height is multiplied by 1.5. If its initial height is 1 meter, find the total height gained over 5 weeks. Answer: 12.1875 m	This activity provides students with guided practice in solving problems related to geometric sequences and series. By working through different scenarios involving nth term calculation, finding unknown terms, and computing the sum of a geometric series, learners will apply key formulas and reinforce their understanding of the structure and behavior of geometric patterns in both abstract and real-life contexts. Additional items be provided. Strategy: Interactive Problem Solving through Collaborative Discussion Instructions: 1. Prepare five activity stations, each with one problem printed on a "Math Chat Card" or projected on a screen. 2. Divide the class into five small groups and assign each group to a station. 3. Provide whiteboards and markers (or digital devices, if available) for each group to show their solutions. 4. Give each group 2-3minutes per station to discuss, solve, and write their answer on their board. 5. After the time is up, each group rotates to the next station until all groups have answered all five problems. 6. After rotations, conduct a class debrief where each group shares their solution to a specific problem and explains their process. 7. Offer scaffolding questions or hints where necessary and emphasize different strategies used. 8. Reinforce the formula for geometric sequences and series during the wrap-up to consolidate learning. Strategy: A Cooperative Math Game Instructions:
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		1. Group the class into teams of 4–5 members each. 2. Set up relay stations in different parts of the room or hallway. Each station contains one problem from the activity (Problems 1–5). 3. Each group will have one “runner” at a time. That student runs to a station, solves the problem (or brings it back), and reports or records the answer. 4. Once the team has a solution, the next runner proceeds to the next station. 5. The group must complete all 5 problems, ensuring accuracy. 6. First team to correctly finish all problems wins a “Math Champ” badge or token. 7. After the game, facilitate a class reflection to go over strategies used, address misconceptions, and reinforce key concepts such as identifying the common ratio, applying the nth term formula, and using the sum formula for geometric series.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: “Load Budgeting Challenge” Scenario: You are given ₱10 mobile load today. Each day, you double your load to make sure you have enough for online classes and social media. Your goal is to decide if this habit is sustainable for 5 days. Instructions: 1. List the amount of load you will have each day if it doubles (Day 1 to Day 5). 2. Use a geometric sequence to calculate: a. The amount of load on Day 5. b. The total load spent in 5 days. 3. Reflect and decide: a. Is this load budget sustainable? b. What can you change to make it more practical?	Activity C.1: “Load Budgeting Challenge” Strategy 1: Interactive Problem-Solving through Role-Playing Instructions: 1. Introduce the Game Setup: 2. Divide the class into teams of 4–5 students. 3. Each team is given the same initial scenario (mobile load budgeting). 4. The class plays through 5 “rounds,” where they calculate the load for each day and answer related questions to “earn” points. Strategy 2: Gameplay Mechanics Instructions: 1. In each round, students calculate the load for a day (starting with Day 1) and record it.

	4. Share your answers in pairs or groups and explain: - What decisions they would make if they had this load pattern in real life. - What strategy would make their load last longer.	2. Teams will calculate the load for Day 5 and the total load used over 5 days using geometric sequence formulas. 3. After each round, teams can "level up" by completing a set of discussion questions (e.g., Is this sustainable? How would you change the strategy?). Team Sharing: At the end of the game, teams will present their strategies and reflect on which budgeting methods they believe are most sustainable. This can include discussing how their approaches would affect their finances over time. Reward System: Points or rewards are given based on correct calculations, group collaboration, and how well students justify their reasoning. This encourages friendly competition and motivates engagement. Strategy 3: Interactive Learning with Digital Tools Instructions: - Provide students with access to the online tool (it could be a simple Google Spreadsheet or an interactive website). - Brief the class on how to use the tool to input their initial values (₱10 for the first day, doubling every day). - Facilitate the guided interaction: - Have students use the tool to calculate the load on each day and find the total load spent over 5 days. - Encourage students to explore how the numbers change as they modify inputs (e.g., changing the initial load or the rate of increase). - After completing the calculations, students will reflect on the sustainability of their mobile load budget.
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		5. Students can modify the parameters (e.g., adjust the doubling rate or initial amount) and observe the new results. 6. Students discuss their findings in groups, exploring what adjustments they would make to keep their budget sustainable. Strategy 4: Group Work with Peer Teaching Instructions: 1. Divide the class into small groups of 3 to 4 students. 2. Distribute a copy of the "Load Budgeting Challenge" to each group and instruct them to calculate the load for each day from Day 1 to Day 5 using geometric sequence formulas. 3. Students work together to solve for the total load spent and discuss whether the load budgeting strategy is sustainable. Encourage them to consider potential adjustments to make the budgeting strategy more practical. 4. Next, have each group develop a strategy to extend their load budget beyond 5 days. They should decide if reducing the doubling rate or limiting usage is necessary and create a short presentation explaining their strategy and reasoning. 5. Once strategies are prepared, each group presents their findings to the class, justifying their decisions and explaining how their strategy would work in real life. 6. After presentations, allow time for a Q&A session where students engage with one another's ideas and further discuss how their strategies could be improved or adjusted.
	C.2. Making Generalization Activity C.2.1: 3 – 2 – 1 Ticket	To wrap-up the lesson facilitates any of the provided activities.

Instructions:
Complete the ticket given.

Three things I learned today:

Two things I found interesting:

One question I still have in mind:

Activity C.2.2: Concept Mapping Activity

- Instructions:
1. Start with a central concept in the middle of your paper or digital canvas:
“Geometric Sequences and Series”
 2. From the center, branch out to the following main nodes:
 - Geometric Sequence
 - Geometric Series
 3. Under Geometric Sequence, create sub-branches for:
 - First Term (a)
 - Common Ratio (r)
 - General Term Formula
 4. Under Geometric Series, create sub-branches for:
 - Finite Series
 - Infinite Series (if applicable)

Activity C.2.1: 3 – 2 – 1 Ticket

Strategy: Active Learning through Self-Assessment

Instructions:

1. Distribute the “3-2-1 Ticket” to the students. Instruct them to complete the ticket with the following:
 - 3 things they learned from the lesson.
 - 2 things they found interesting or noteworthy.
 - 1 question they still have or need clarification on.
2. Once the tickets are completed, collect them to gauge the class’s overall understanding. Use the responses to guide further discussion, address misconceptions, or plan the next steps in your teaching. This method can be adapted for any subject matter and is useful for closing a lesson with student engagement.

Strategy: *Real-time Feedback through Digital Tools*

Instructions:

1. Set up a digital form or poll where students submit their 3-2-1 responses.
2. After students complete the form, review the responses and share a summary of the key takeaways with the class. You can also use this data for follow-up discussions, clarifications, or in future lessons.

Activity C.2.2: Concept Mapping Activity

Strategy: Visual Learning and Concept Organization

Instructions:

1. Provide students with the necessary tools to create their concept maps, whether it’s paper and markers or digital tools like Mind Meister or Google Drawings.
2. Instruct them to start with the central concept “Geometric Sequences and Series” and create branches for the related sub-concepts. Encourage students to

	5. Include examples, illustrations, or real-life applications on the side branches (e.g., bacterial growth, savings accounts, cellphone load doubling, etc.) 6. Highlight connections between terms and concepts using arrows, colors, or labels to show how ideas relate.	incorporate real-life applications, examples, or visual representations. 3. After they complete their concept maps, have them share their work in pairs or small groups. Facilitate a discussion to clarify connections between ideas and ensure that students have accurately grasped the concepts. Strategy: Interactive Concept Mapping in Groups Instructions: 1. Assign students to small groups and have them collaboratively create a digital concept map on a shared platform. 2. Each group will map out the concepts related to geometric sequences and series, adding examples and making connections. 3. After completion, students will present their maps to the class, explaining the connections they made between concepts.
	<i>C.3. Evaluating Learning</i> Activity C.3: Assessing Learners Understanding on Geometric Sequence and Series Instructions: Do as indicated. <i>Part I. Multiple Choices</i> Choose the letter of the best answer. 1. Which of the following is a geometric sequence? A. 2, 4, 6, 8, ... B. 3, 6, 12, 24, ... C. 1, 3, 6, 10, ... D. 10, 9, 8, 7, ...	Activity C.3: Assessing Learners Understanding on Geometric Sequence and Series Instructions: 1. Facilitate the evaluation by distributing the printed or digital activity. 2. Remind students to answer all parts: multiple choice, problem solving, and reflection. 3. Monitor and assist struggling students, especially in problem solving. 4. After completion, conduct a feedback session where learners discuss answers and reflect on errors and learnings.

	2. What is the common ratio in the geometric sequence 5, 15, 45, 135, ...? A. 3 B. 10 C. 15 D. 5 3. In the geometric sequence, what does the variable a_1 represent? A. nth term B. number of term C. first term D. sum of terms 4. What does a geometric series represent? A. A list of repeated numbers B. A pattern that increases by addition C. The sum of terms in a geometric sequence D. The difference between terms in a pattern 5. Which formula is used to find the sum of the first n terms in a geometric series when $r \neq 1$. A. $S_n = a_1 \frac{1-r^n}{1-r}$ B. $S_n = \frac{a_1}{1-r}$ C. $S_n = a_1 r^{n-1}$ D. $S_n = \frac{n}{2}(a_1 + a_n)$ <i>Part II: Problem Solving</i> Solve and show your complete solution. 1. Geometric Sequence Jorge saves money by doubling his savings every week. In the first week, he saves ₱50. How much will he have saved in total by the 6th week? 2. Geometric Series A ball is dropped and it bounces back to $\frac{3}{4}$ of its previous height. If the initial drop is 16 meters, what is the total vertical distance the ball travels at the 4th bounce?	Strategy: Kahoot or Quizizz-Based Assessment Instructions: 1. Create a quiz using platforms like Kahoot or Quizizz based on Part I. 2. Display questions one at a time and explain correct answers afterward. 3. For problem-solving and essay parts, assign follow-up tasks using Google Docs or paper to maintain reflective and computational aspects.
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	Part III: Reflective Essay. Answer the question below in 3–5 sentences. <i>Why do you think understanding geometric sequences and series is important in real-life situations like savings, investments, or technology? Give one example where this concept could help you in the future.</i>	
	C.4. Additional Activities Activity C.4: More Exercises on Geometric Sequence and Series <i>Remediation</i> Problem 1: Identifying the Common Ratio Given: 2, 6, 18, 54, ... Question: What is the common ratio of the geometric sequence? Problem 2: Finding the nth Term Given: The geometric sequence is 5, 10, 20, 40, ... Question: What is the 6th term? Problem 3: Finding the Sum of a Geometric Series Given: Find the sum of the first 4 terms of the geometric series: $3 + 6 + 12 + 24 + \dots$ <i>Enhancement</i> Problem 1: Finding the Term Position (n) Given: In a geometric sequence, the first term is 2 and the common ratio is 3. Question: Which term is equal to 486? Problem 2: Word Problem – Real-Life Context Scenario: A small sari-sari store sells bottled water. On day 1, it sold 2 bottles. Each day, the number of bottles	Activity C.4: More Exercises on Geometric Sequence and Series This supplementary activity may be assigned as homework. For in-class support, small group remediation should be provided to learners who have not yet mastered the lesson. Each group should consist of a mix of high-performing, average, and low-performing learners to encourage peer support and collaborative learning. Meanwhile, enrichment activities should be offered to those who are ready for more advanced tasks to further deepen their understanding. Provide additional items if necessary.

	sold is tripled. Question: How many bottles were sold on the 7th day? Problem 3: Sum of Geometric Series Given: A geometric series has a first term of 8 and a common ratio of $\frac{1}{2}$. Question: What is the sum of the infinite geometric series?	
VI. ASSESSMENTS	Please see the attached unit test and performance task.	
VII. REFLECTION	For the teachers, you may answer the following questions: 1. What went well in your teaching? 2. What challenges did you encounter? 3. How did you manage these challenges?	

UNIT TEST

Directions: Choose the letter of the best answer. Write your answers on the answer sheet provided.

- Which sequence is closely linked to the number of flower petals?
A. Arithmetic B. Geometric
C. **Fibonacci**
D. Prime numbers
- In the sequence 3, 9, 27, 81..., what is the pattern?
A. Multiply by 2
B. Add 6
C. **Multiply by 3**
D. Subtract 3
- In the sequence 3, 9, 15, 21..., what is the pattern?
A. Multiply by 2
B. **Add 6**
C. Multiply by 3
D. Subtract 3
- Which sequence is arithmetic?
A. 1, 2, 4, 8
B. **2, 5, 8, 11**
C. 3, 6, 12, 24
D. 5, 10, 20, 40
- Which of the following is a geometric sequence?
A. 1, 4, 7, 10
B. **2, 4, 8, 16**
C. 3, 6, 9, 12
D. 10, 20, 30, 40
- What is the common ratio in the geometric sequence 3, 6, 12, 24...?
A. 3
B. 6
C. **2**
D. 1
- What is the 10th term in the sequence 3, 6, 9, 12...?
A. **30**
B. 33
C. 36
D. 39
- Find the 5th term of the sequence: 2, 6, 18, ...
A. 54
B. 108
C. **162**
D. 486
- What is the common difference if the 1st term is 7 and the 4th term is 19?
A. 3
B. **4**
C. 5
D. 6
- Which formula gives the nth term of an arithmetic sequence?
A. $a_n = a_1 \times r^{n-1}$
B. **$a_n = a_1 + (n - 1)d$**
C. $a_n = a_1 - d(n - 1)$
D. $a_n = a_1 \div n$
- What distinguishes a sequence from a series?
A. A series has no pattern
B. A sequence is a sum

- C. A series is the sum of the terms of a sequence
 D. A sequence has letters
12. Which of the following is a sequence?
 A. $2 + 4 + 6 + 8$
 B. 2, 4, 6, 8
 C. 20
 D. $\sum 2n$
13. What does $\sum_{i=1}^4 i$ equal?
 A. 10
 B. 5
 C. 15
 D. 20
14. What is the sigma notation for the series $2 + 4 + 6 + 8$?
 A. $\sum_{n=1}^4 2n$
 B. $\sum_{n=1}^4 n$
 C. $\sum_{n=1}^4 4n$
 D. $\sum_{n=1}^4 (2n - 1)$
15. Evaluate the sum: $\sum_{k=1}^3 3k$
 A. 6
 B. 9
 C. 12
 D. 18
16. What is the sum of the geometric series: $2 + 4 + 8 + 16$?
 A. 30
 B. 31
 C. 32
 D. 33
17. How many terms are in the arithmetic series if the first term is 4, last term is 40, and the common difference is 4?
 A. 8
 B. 9
 C. 10
 D. 11
18. Find the sum of the first 6 terms of a geometric sequence where $a_1 = 1$ and $r = 2$
 A. 63
 B. 64
 C. 65
 D. 66
19. Solve: What is the sum of the series $\sum_{n=1}^5 i^2$
 A. 30
 B. 35
 C. 55
 D. 65
20. A ball bounces to $\frac{1}{2}$ of its height each time. If dropped from 16 m, what is the total distance traveled by the ball when it hits the ground for the 3rd time (down + up)?
 A. 28
 B. 40
 C. 56
 D. 60

PERFORMANCE TASK

Task Name: InvestiMath: Using Sequences to Grow Wealth

Goal: Demonstrate how arithmetic and geometric sequences and series can be used to make sound financial decisions in real-life scenarios such as saving, investing, or managing a business.

Role: You are a financial planner or business owner. You are either creating a personal savings plan, projecting business income/expenses, or analyzing loan or investment growth using sequences and series.

Audience: Your audience includes your teacher (as an investor or client), classmates, and possibly community members interested in financial literacy.

Situation: You are tasked with creating a financial plan or proposal that uses arithmetic and/or geometric sequences and series to solve a real-life financial problem. Choose one of the following scenarios:

Scenario 1: Saving for a New Bike

You want to buy a bicycle worth ₱6,000. You decide to save ₱500 every month from your allowance. Create a savings plan showing how your savings will grow each month. Use an arithmetic sequence to represent your savings and determine how many months it will take to buy the bike.

Scenario 2: Growing Ice Cream Sales

You start a small ice cream cart business. On your first day, you earn ₱200. Every day, your sales increase by ₱50 because of word of mouth and regular customers. Create a projection of your total earnings over the first 7 days using an arithmetic series. How much money will you have earned by the end of the week?

Scenario 3: Multiplying Your Coin Savings

You decide to save coins by increasing the amount you save each day for 30 days. You start with ₱1 on the first day, ₱5 on the second day, ₱25 on the third day, and so on. Represent your daily savings using a geometric sequence. Calculate how much you will have saved in total by the end of the 30 days.

Product/Performance: Create a financial report or visual presentation (poster, slides, or infographic) that includes:

- Mathematical modeling of the situation using correct formulas.
- Calculations showing your solution process.
- Graphs that illustrate the data.
- A short-written reflection explaining how sequences and series helped you make an informed financial decision.

Standards: Your work will be assessed based on the following criteria:

Rubric for Scoring:

Criteria	4 – Exemplary	3 – Proficient	2 – Developing	1 – Beginning
Mathematical Accuracy (30%)	Accurately applies arithmetic and/or geometric sequence and series formulas with no errors in reasoning or calculations.	Applies formulas correctly with minor calculation or reasoning errors.	Shows basic understanding but includes significant calculation or conceptual errors.	Misapplies formulas or demonstrates little understanding of sequences and series.

Real-Life Application and Analysis (10%)	Clearly and insightfully connects mathematical models to a realistic financial scenario; shows depth in financial reasoning.	Connects mathematical models to real-life scenarios with adequate explanation.	Attempts a connection, but explanation lacks depth or contains inconsistencies.	Connection to real-life applications is unclear, inaccurate, or missing.
Solution Process and Justification (30%)	All steps in the solution are logical, clearly explained, and well-organized; justification is complete and compelling.	Most steps are clearly shown and explained; justification is provided but may lack clarity or detail.	Some steps are missing or unclear; justification is minimal.	Solution process is incomplete or confusing; no meaningful justification.
Visual Presentation and Communication (10%)	Presentation is highly organized, visually appealing, and enhances understanding with effective use of graphs and visuals.	Presentation is organized and mostly clear; graphs and visuals are included and mostly accurate.	Presentation is somewhat disorganized or unclear; visuals/graphs may be incomplete or contain errors.	Presentation lacks clarity or visuals; graphs may be missing or not relevant.
Reflection and Financial Insight (10%)	Reflection shows strong insight into the usefulness of sequences and series in making financial decisions; well-articulated.	Reflection discusses how sequences and series helped, with adequate clarity and thought.	Reflection is brief or somewhat superficial; lacks clear connection to mathematical ideas.	Reflection is missing, unclear, or lacks meaningful insight.